

Study Guide

Chapters 0–5: numbers, logic, sets, functions, induction, relations

Theory you need for the **essential** and **recommended** exercises of the planning, with worked examples in the style the course expects.

Book: C. Newstead, *An Infinite Descent into Pure Mathematics*. Section numbers (e.g. §1.3, Theorem 5.2.35, exercise 0.E.22) refer to that book.

Covers: Ch. 0 (numbers, number bases, division, irrationals), §1.1–1.3, §2.1–2.2, §3.1–3.2, §4.2–4.3, §5.1–5.2 and the chapter exercises 0.E–5.E named in the planning.

Quizzes: Quiz 1 (23 Sep): numbers + propositional logic. Quiz 2 (7 Oct): sets + functions. Quiz 3 (21 Oct): relations + modular arithmetic. Exam 4 Nov (70 %).

How to use this guide

Every chapter of this guide is one chapter of the book, and every lettered section (**A**, **B**, ...) is one book section in the order of the planning. Each chapter runs up a ladder of three difficulty levels:

Definition / key fact

Learn these word for word. Almost every exercise starts with “unfold the definition”.

Theorem

Results you may quote in a proof (with their name or number).

Proof strategy

The template to copy. The course grades the *structure* of a proof as much as the maths.

Example Self Tutor

Level 1. One proof shape at a time, written the way you should write it on a quiz. Cover the solution and try first.

Example Book level

Level 2. The difficulty of the hardest essential and recommended exercises: several shapes in one proof, abstract objects, and an *invention step* that is named so you learn to spot it.

STRETCH EXERCISES

Level 3. Unsolved problems with hints, at the top of the book’s range. Full solutions in Appendix S. Do these before the exam, not before the quiz.

EXERCISES

Which planning exercises the section prepares you for, and the skills they test.

Watch out

The mistakes that cost the most marks.

Suggested order for each section: read the boxes, do the Level 1 examples with the solution covered, then the planning’s essential exercises; then the Level 2 examples and the recommended exercises; then the stretch exercises when preparing for the exam.

Conventions used in the book and in this guide

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$: **zero is a natural number** in this book. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ as usual.
- A set is **inhabited** if it has at least one element (the book’s word for “non-empty”).
- $X \subseteq Y$ means subset (equality allowed); $X \subsetneq Y$ means proper subset.

- $f[U]$ is the image of $U \subseteq X$ and $f^{-1}[V]$ the preimage of $V \subseteq Y$ (square brackets; $f^{-1}[V]$ exists even when f has no inverse).
- $[a]_{\sim}$ or $[a]$ is the equivalence class of a ; $[a]_n$ is the class of a in $\mathbb{Z}/n\mathbb{Z}$.
- Logical symbols: $\wedge$ and, $\vee$ or, $\Rightarrow$ implies, $\Leftrightarrow$ iff, $\neg$ not, $\forall$ for all, $\exists$ there exists, $\exists!$ there exists a unique.
- $\square$ marks the end of a proof. {grey text in braces} is the justification of a step.

Planning at a glance (chapters 0–5)

Date	Sections	Essential	Recommended
Mon 7 Sep	Ch. 0 (to irrationals), A.1	Ch0: 2, 21, 24, 40 0.E: 3, 4, 22, 27, 28	Ch0: 14, 30, 38, 45, 51 0.E: 17, 18, 29, 30
Mon 14 Sep	§1.1, §1.3 (to ex. 1.3.28)	1.1: 21, 25, 32, 33, 38, 47, 56, 57 1.3: 5, 13, 14, 22, 27	1.1: 37, 43, 54, 55, 65, 66 1.E: 11
Wed 16 Sep	§1.2, §1.3 (from thm 1.3.29)	1.2: 6, 15, 18, 39 1.3: 44ace; 1.E: 18	1.2: 21, 22, 28, 37ab 1.3: 32, 35; 1.E: 28, 29, 30
Mon 21 Sep	§2.1, §2.2	2.1: 10, 27, 31, 35 2.2: 5, 24, 26, 28, 35, 36, 37, 40, 51 2.E: 6, 33, 38	2.1: 17, 18, 23, 24, 32, 36 2.2: 4, 6, 9, 25, 41, 45, 53 2.E: 4, 5, 16, 28, 32, 41
Wed 23 Sep	Quiz 1 ; §2.2, §3.1	3.1: 11, 15abc, 36 3.E: 13, 17, 19, 34, 41	3.1: 8, 16, 19, 33, 40, 41 3.E: 2, 37, 38
Mon 28 Sep	§3.2	3.2: 5, 8, 12, 14, 20, 40, 47 3.E: 24, 25, 48	3.2: 6, 16, 19, 25, 39ab, 45, 46 3.E: 35, 47
Wed 30 Sep	§4.2	4.2: 5, 11 4.E: 10, 11, 16, 17, 18, 20, 25	4.2: 8, 17, 20 4.E: 12, 14, 15, 21, 24, 33
Mon 5 Oct	§4.3	4.3: 8, 9, 14, 15 4.E: 13, 19, 26, 27, 29, 30, 31, 32	4.3: 10, 18 4.E: 28, 34, 35
Wed 7 Oct	Quiz 2 ; §5.1, §5.2	5.1: 6, 13, 22, 23, 28, 39, 40, 41 5.2: 24, 27, 33; 5.E: 18, 22, 23	5.1: 10, 11, 18, 24, 29, 33, 34 5.2: 25, 28, 30, 34, 36

Writing a proof (Appendix A.1 in one box)

The six habits of a readable proof

- 1 Say what you are proving and in which form.** “We prove that for all $n \in \mathbb{Z}$, if n is odd then n^2 is odd.” Identify the outermost logical shape ($\forall$, $\Rightarrow$, $\wedge$, $\dots$): that shape decides the first line of the proof (Chapter 1).
- 2 Introduce every variable before you use it.** “Let $n \in \mathbb{Z}$.” “Fix $k \in \mathbb{Z}$ such that $n = 2k + 1$.” A letter that appears from nowhere is an error.
- 3 Unfold definitions.** “ n is odd” means “ $n = 2k + 1$ for some $k \in \mathbb{Z}$ ”. Most exercises are solved by unfolding the definitions on both sides and connecting them.
- 4 Justify each step** with “by definition of ...”, “by assumption”, “by Theorem ...”, or a one-line calculation. Never write “clearly”.
- 5 Write sentences.** $\Rightarrow$ and $\therefore$ are not abbreviations for “so”. Chains of equations are fine; chains of unexplained symbols are not.
- 6 Close the loop.** End with the statement you set out to prove and a $\square$. If the proof had cases, say that all cases are covered.

Contents

0	Getting started: numbers	4
A	Number sets and set-builder notation	4
B	Number bases	5
C	Division of integers	5
D	Rational and irrational numbers	7
E	Book-level examples	8
F	Stretch exercises	10
G	Review set 0	10
1	Logical structure	11
A	Propositional logic	11
B	Variables and quantifiers	13
C	Logical equivalence	15
D	Book-level examples	18
E	Stretch exercises	20
F	Review set 1	20
2	Sets	21
A	Sets, subsets and the power set	21
B	Set operations	22
C	Book-level examples	26
D	Stretch exercises	28
E	Review set 2	28
3	Functions	29
A	Functions, images and preimages	29
B	Injections, surjections and bijections	31
C	Book-level examples	34
D	Stretch exercises	35
E	Review set 3	36
4	Mathematical induction	37
A	Weak induction	37
B	Strong induction and well-ordering	39
C	Book-level examples	41
D	Stretch exercises	43
E	Review set 4	44

5	Relations	45
A	Relations and their properties	45
B	Equivalence relations and partitions	47
C	Book-level examples	50
D	Stretch exercises	52
E	Review set 5	52
T	One-page proof toolbox	53
A	Reading off the first line of a proof	53
S	Solutions to the stretch exercises	54
A	Chapter 0	54
B	Chapter 1	54
C	Chapter 2	55
D	Chapter 3	56
E	Chapter 4	56
F	Chapter 5	57

Chapter 0

Getting started: numbers

Contents

- A** Number sets and set-builder notation
- B** Number bases
- C** Division of integers
- D** Rational and irrational numbers
- E** Book-level examples
- F** Stretch exercises
- G** Review set 0

Planning

Essential: Ch0: 2, 21, 24, 40; 0.E: 3, 4, 22, 27, 28

Recommended: Ch0: 14, 30, 38, 45, 51; 0.E: 17, 18, 29, 30
Tested in **Quiz 1** (23 Sep) together with §1.1.

Chapter 0 is where the book introduces the objects you will prove things about for the rest of the course, and where you see the first real proofs. The logic behind these proofs is only explained in Chapter 1, so in this chapter copy the *shape* of the model solutions; by Chapter 1 you will know why the shape is right.

A NUMBER SETS AND SET-BUILDER NOTATION

The number sets

$$\mathbb{N} = \{0, 1, 2, 3, \dots\} \subseteq \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}, \quad \mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

In this book $0 \in \mathbb{N}$. Writing $n \geq 1$ or $n \in \mathbb{N}, n \neq 0$ when you need a positive natural number is your job.

Set-builder notation

There are two forms, and you must be able to read both:

- **Filter form** $\{x \in X \mid p(x)\}$: the elements of X that satisfy the condition $p(x)$.
Reading rule: $a \in \{x \in X \mid p(x)\} \Leftrightarrow a \in X$ and $p(a)$.
- **Image form** $\{e(x) \mid x \in X\}$: all values of the expression $e(x)$ as x runs through X .
Reading rule: $b \in \{e(x) \mid x \in X\} \Leftrightarrow b = e(x)$ for some $x \in X$.

Intervals are filter sets: $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$, $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$, $[a, \infty) = \{x \in \mathbb{R} \mid x \geq a\}$.

The two reading rules are the whole secret of Chapter 2: to decide whether something is in a set, replace the set by its condition.

Example 1 — reading set-builder notation

Self Tutor

- 1 Write the set E of even integers in filter form and in image form.
- 2 List the elements of $\{n \in \mathbb{N} \mid n^2 < 20\}$.
- 3 Is $7 \in \{3k + 1 \mid k \in \mathbb{Z}\}$? Is 8?

$$1 \quad E = \{n \in \mathbb{Z} \mid n \text{ is even}\} = \{n \in \mathbb{Z} \mid \exists k \in \mathbb{Z}, n = 2k\} = \{2k \mid k \in \mathbb{Z}\}.$$

- 2 $n^2 < 20$ holds for $n = 0, 1, 2, 3, 4$ and fails from $n = 5$ on, so the set is $\{0, 1, 2, 3, 4\}$. $\{0 \in \mathbb{N}!\}$
- 3 $7 = 3 \cdot 2 + 1$ with $2 \in \mathbb{Z}$, so yes. For 8 we would need $3k + 1 = 8$, i.e. $k = \frac{7}{3} \notin \mathbb{Z}$, so $8 \notin \{3k + 1 \mid k \in \mathbb{Z}\}$.
□

B NUMBER BASES

Base- b expansion

Let $b \in \mathbb{N}$ with $b \geq 2$. A **base- b expansion** of $n \in \mathbb{N}$ is an expression

$$n = d_r b^r + d_{r-1} b^{r-1} + \cdots + d_1 b + d_0, \quad d_i \in \{0, 1, \dots, b-1\},$$

written $n = (d_r d_{r-1} \dots d_1 d_0)_b$. The d_i are the **base- b digits**. Base 2 = binary, base 10 = decimal, base 16 = hexadecimal (digits $0, \dots, 9, A, \dots, F$).

Theorem (existence and uniqueness of base- b expansions)

Let $b \geq 2$. Every natural number $n \geq 1$ has a base- b expansion, and the expansion is unique once leading zeros are discarded ($d_r \neq 0$).

Existence is proved by strong induction in Chapter 4 (see Example 4.11); uniqueness uses the division theorem of Section C. For the exercises you need the *algorithm* and the ability to *use* the expansion in a proof.

Converting between bases

Decimal $\rightarrow$ base b : divide by b , write down the remainder, divide the quotient by b , ... until the quotient is 0. The remainders, read *last to first*, are the digits.

Base $b \rightarrow$ decimal: evaluate $d_r b^r + \cdots + d_0$.

Check by converting back.

Example 2 — conversions

Self Tutor

(a) Write 2026 in base 7. (b) Write $(10110)_2$ and $(2F)_{16}$ in decimal.

(a) $2026 = 7 \cdot 289 + 3$, $289 = 7 \cdot 41 + 2$, $41 = 7 \cdot 5 + 6$, $5 = 7 \cdot 0 + 5$.

Reading the remainders upwards: $2026 = (5623)_7$. {check: $5 \cdot 343 + 6 \cdot 49 + 2 \cdot 7 + 3 = 1715 + 294 + 14 + 3 = 2026$ ✓}

(b) $(10110)_2 = 1 \cdot 16 + 0 \cdot 8 + 1 \cdot 4 + 1 \cdot 2 + 0 = 22$; $(2F)_{16} = 2 \cdot 16 + 15 = 47$. □

Proofs about digits always start the same way: write $n = 10m + d_0$ (or $n = bm + d_0$), where $m = (d_r \dots d_1)_b$ is the number formed by the other digits. See Example 6 after we have divisibility.

C DIVISION OF INTEGERS

Division theorem

Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There exist *unique* $q, r \in \mathbb{Z}$ such that

$$a = qb + r \quad \text{and} \quad 0 \leq r < |b|.$$

q is the **quotient** and r the **remainder** of a on division by b .

The remainder is never negative: $-7 = (-2) \cdot 4 + 1$, so -7 has quotient -2 and remainder 1 on division by 4 (not -3).

Divisibility, even and odd, prime

Let $a, b \in \mathbb{Z}$.

- b **divides** a , written $b \mid a$, if there exists $q \in \mathbb{Z}$ with $a = qb$. (Equivalently: the remainder of a on division by b is 0 .) We also say a is a **multiple** of b and b is a **divisor** of a . Note $b \mid 0$ for every b , and $1 \mid a$ for every a .
- a is **even** if $2 \mid a$, i.e. $a = 2k$ for some $k \in \mathbb{Z}$; a is **odd** if $a = 2k + 1$ for some $k \in \mathbb{Z}$.
- **Fact** (division theorem with $b = 2$): every integer is even or odd, and not both.
- $p \in \mathbb{N}$ is **prime** if $p \geq 2$ and the only positive divisors of p are 1 and p .

Working with $\mid$ and parity

- **To prove $b \mid a$** : produce an explicit integer q and check $a = qb$. Say why $q \in \mathbb{Z}$.
- **To use an assumption $b \mid a$** : “Fix $q \in \mathbb{Z}$ with $a = qb$.” Then compute with qb instead of a .
- **To prove something about an arbitrary integer n** : often split into cases $n = 2k$ and $n = 2k + 1$ (or $n = 3k, 3k + 1, 3k + 2, \dots$) using the division theorem. Make sure the cases cover everything.
- Never write $\frac{a}{b} \in \mathbb{Z}$ as a definition of divisibility: $\frac{a}{b}$ is a rational number, and the exercises want the integer q .

Example 3 — using and proving divisibility**Self Tutor**

Let $a, b, c \in \mathbb{Z}$. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$.

Suppose $a \mid b$ and $a \mid c$. Fix $k, l \in \mathbb{Z}$ with $b = ka$ and $c = la$. {definition of $\mid$ }

Let $x, y \in \mathbb{Z}$. Then

$$bx + cy = kax + lay = (kx + ly)a.$$

Since $kx + ly \in \mathbb{Z}$, this shows $a \mid (bx + cy)$ by definition of divisibility. □

Example 4 — proof by cases on the remainder**Self Tutor**

Let $n \in \mathbb{Z}$. Prove that the remainder of n^2 on division by 4 is 0 or 1 .

By the division theorem, $n = 2k$ or $n = 2k + 1$ for some $k \in \mathbb{Z}$.

- If $n = 2k$, then $n^2 = 4k^2 = 4k^2 + 0$, so the remainder is 0 . { $0 \leq 0 < 4$ }
- If $n = 2k + 1$, then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$, so the remainder is 1 . { $0 \leq 1 < 4$ }

In both cases the remainder is 0 or 1 . {uniqueness of the remainder in the division theorem} □

Example 5 — digits and divisibility

Self Tutor

Prove that a natural number n is even if and only if its last decimal digit is even.

Write $n = (d_r \dots d_1 d_0)_{10}$ and let $m = (d_r \dots d_1)_{10}$, so that $n = 10m + d_0$ with $0 \leq d_0 \leq 9$.

($\Rightarrow$) Suppose n is even, say $n = 2l$ with $l \in \mathbb{Z}$. Then $d_0 = n - 10m = 2l - 10m = 2(l - 5m)$, so d_0 is even.

($\Leftarrow$) Suppose d_0 is even, say $d_0 = 2j$. Then $n = 10m + 2j = 2(5m + j)$, so n is even. $\square$

The same trick (“ $n = bm + d_0$ ”) proves: n is even iff its last binary digit is 0; $5 \mid n$ iff the last decimal digit is 0 or 5. For “ $3 \mid n$ iff $3 \mid$ digit sum” you also need $10^k = 9(\dots) + 1$, which comes from induction (Chapter 4).

D RATIONAL AND IRRATIONAL NUMBERS**Rational and irrational**

A real number x is **rational** if there exist $a, b \in \mathbb{Z}$ with $b \neq 0$ and $x = \frac{a}{b}$; otherwise x is **irrational**.

Lowest terms. Every rational number can be written as $\frac{a}{b}$ with $b > 0$ and a, b having no common divisor greater than 1; in particular a and b are not both even. (Divide out common factors; the process stops because b decreases.)

Because “irrational” is a *negative* statement (“there are no such a, b ”), irrationality is always proved by **contradiction**: assume $x = \frac{a}{b}$ in lowest terms and derive something impossible. The lemma that makes it work:

Lemma

Let $n \in \mathbb{Z}$. If n^2 is even, then n is even.

Proof. We prove the contrapositive: if n is odd then n^2 is odd. Suppose $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd. $\square$ (Why the contrapositive is allowed: §1.3.)

Theorem: $\sqrt{2}$ is irrational

Proof. Suppose, for a contradiction, that $\sqrt{2} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b > 0$, not both even. Squaring, $2b^2 = a^2$, so a^2 is even, so a is even by the Lemma; write $a = 2c$. Then $2b^2 = 4c^2$, so $b^2 = 2c^2$ is even, so b is even. Now a and b are both even, contradicting the choice of a, b . Hence $\sqrt{2}$ is irrational. $\square$

Example 6 — $\sqrt{3}$ is irrational

Self Tutor

Prove that $\sqrt{3}$ is irrational.

Step 1 (lemma). If $n \in \mathbb{Z}$ and $3 \mid n^2$ then $3 \mid n$. Contrapositive: suppose $3 \nmid n$. Then $n = 3k + 1$ or $n = 3k + 2$. {division theorem}

If $n = 3k + 1$: $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$; if $n = 3k + 2$: $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. In both cases n^2 has remainder 1, so $3 \nmid n^2$.

Step 2. Suppose $\sqrt{3} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b > 0$, no common divisor > 1 . Then $3b^2 = a^2$, so $3 \mid a^2$, so $3 \mid a$ by Step 1; write $a = 3c$. Then $3b^2 = 9c^2$, so $b^2 = 3c^2$, so $3 \mid b^2$, so $3 \mid b$. Now 3 divides both a and b : contradiction. Hence $\sqrt{3}$ is irrational. $\square$

Example 7 — irrational plus rational

Self Tutor

Let $x \in \mathbb{R}$ be irrational and $r \in \mathbb{Q}$. Prove that $x + r$ is irrational, and that rx is irrational if $r \neq 0$.

Suppose, for a contradiction, that $x + r$ is rational, say $x + r = \frac{a}{b}$; write $r = \frac{c}{d}$ ($b, d \neq 0$). Then

$$x = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} \in \mathbb{Q}, \quad \{ad - bc, bd \in \mathbb{Z} \text{ and } bd \neq 0\}$$

contradicting that x is irrational. So $x + r$ is irrational.

Similarly, if $r \neq 0$ and $rx = \frac{a}{b}$, then $x = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc} \in \mathbb{Q}$ (note $c \neq 0$ because $r \neq 0$): contradiction. $\square$

Example 8 — parity contradiction

Self Tutor

Prove that $\log_2 3$ is irrational.

Note $\log_2 3 > 0$. Suppose $\log_2 3 = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b > 0$; then $a > 0$ too. By definition of log, $2^{a/b} = 3$, so $2^a = 3^b$. But 2^a is even (since $a \geq 1$) and 3^b is odd (a product of odd numbers is odd, Example 1.4), so an even number equals an odd number: contradiction. $\square$

Watch out

- The irrationals are *not* closed under $+$ or $\times$: $\sqrt{2} + (-\sqrt{2}) = 0$ and $\sqrt{2} \cdot \sqrt{2} = 2$. “Irrational times irrational is irrational” is false; do not use it.
- In a lowest-terms argument, say explicitly which property you contradict (“not both even”, “no common divisor > 1 ”). Without that sentence the contradiction is not there.
- $0 \in \mathbb{N}$: the base case of statements “for all $n \in \mathbb{N}$ ” is $n = 0$, and $n^2 < 20$ has the solution $n = 0$.

EXERCISES Chapter 0 and 0.E

Essential: Ch0: 2, 21, 24, 40 · 0.E: 3, 4, 22, 27, 28

Recommended: Ch0: 14, 30, 38, 45, 51 · 0.E: 17, 18, 29, 30

Skills they test: reading and writing set-builder notation (A); converting numbers between bases and proving statements about digits (B); proving $b \mid a$ from the definition, parity arguments, splitting into cases by remainder (C); lowest-terms contradiction proofs of irrationality and “rational $\pm$ irrational” arguments (D). Exercise 45 is the last division exercise you need.

E BOOK-LEVEL EXAMPLES

The examples in Sections A–D each show *one* proof shape. The hardest exercises in the book’s Chapter 0 combine two or three shapes and ask you to *invent* something: a witness, a factorisation, the right remainder to look at. Each example below names its invention step. Cover the solution, find the invention yourself, then compare.

Example 9 — $\sqrt{2} + \sqrt{3}$ is irrational

Book level

Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Invention: isolate a number you already know is irrational.

Suppose, for a contradiction, that $x = \sqrt{2} + \sqrt{3}$ is rational. Note $x > 0$, so $x \neq 0$. Then $x - \sqrt{2} = \sqrt{3}$;

squaring both sides,

$$x^2 - 2\sqrt{2}x + 2 = 3, \quad \text{so} \quad \sqrt{2} = \frac{x^2 - 1}{2x}.$$

Since $x \in \mathbb{Q}$ and $x \neq 0$, the right-hand side is rational (sums, differences, products and quotients by non-zero numbers of rationals are rational). Hence $\sqrt{2} \in \mathbb{Q}$, contradicting the Theorem in Section D. So $\sqrt{2} + \sqrt{3}$ is irrational. $\square$

Example 10 — sums of two squares

Book level

Prove that no integer of the form $4k + 3$ (with $k \in \mathbb{Z}$) is a sum of two squares of integers.

Invention: compare remainders on division by 4, using the uniqueness clause of the division theorem.

Suppose, for a contradiction, that $4k + 3 = a^2 + b^2$ with $a, b, k \in \mathbb{Z}$. By Example 0.4, $a^2 = 4s + r$ and $b^2 = 4t + r'$ for some $s, t \in \mathbb{Z}$ and $r, r' \in \{0, 1\}$. Then

$$a^2 + b^2 = 4(s + t) + (r + r'), \quad 0 \leq r + r' \leq 2 < 4,$$

so by uniqueness of quotient and remainder, the remainder of $a^2 + b^2$ on division by 4 is $r + r' \in \{0, 1, 2\}$. But $4k + 3$ has remainder 3. Contradiction. $\square$

Example 11 — divisibility by 9 and digit sums

Book level

For $n \in \mathbb{N}$ let $s(n)$ be the sum of the decimal digits of n . Prove that $9 \mid n$ if and only if $9 \mid s(n)$.

Invention: prove the stronger fact $9 \mid n - s(n)$; then both directions are one line each.

Step 1: $9 \mid 10^k - 1$ for every $k \in \mathbb{N}$. For $k = 0$, $10^0 - 1 = 0 = 9 \cdot 0$. For $k \geq 1$, expanding the right-hand side of

$$10^k - 1 = (10 - 1)(10^{k-1} + 10^{k-2} + \cdots + 10 + 1)$$

shows the identity (all middle terms cancel), so $10^k - 1 = 9m_k$ with $m_k \in \mathbb{N}$.

Step 2. Write $n = \sum_{i=0}^r d_i 10^i$ with digits d_i . Then

$$n - s(n) = \sum_{i=0}^r d_i 10^i - \sum_{i=0}^r d_i = \sum_{i=0}^r d_i (10^i - 1) = 9 \sum_{i=0}^r d_i m_i,$$

so $9 \mid n - s(n)$.

Step 3. If $9 \mid n$ then $9 \mid n - (n - s(n)) = s(n)$ by Example 0.3. If $9 \mid s(n)$ then $9 \mid s(n) + (n - s(n)) = n$. $\square$

Since $3 \mid 9m_k$, the same proof gives the rule for 3. Replacing $10^k - 1$ by $10^k - (-1)^k$ gives the alternating-sum rule for 11.

Example 12 — an irrational between any two rationals

Book level

Let $p, q \in \mathbb{Q}$ with $p < q$. Prove that there is an irrational number x with $p < x < q$.

Invention: scale a known irrational so that it fits in the gap.

Put $x = p + \frac{q-p}{\sqrt{2}}$.

x is irrational. $\frac{1}{\sqrt{2}} = \frac{1}{2} \cdot \sqrt{2}$ is irrational by Example 0.7 (non-zero rational times irrational). Then $(q-p) \cdot \frac{1}{\sqrt{2}}$ is irrational by the same example, since $q-p$ is a non-zero rational. Then $x = p + (\text{irrational})$ is irrational, again by that example.

x lies between p and q . Since $q-p > 0$ and $0 < \frac{1}{\sqrt{2}} < 1$, we have $0 < \frac{q-p}{\sqrt{2}} < q-p$, hence $p < x < q$. $\square$

Example 13 — uniqueness of base- b expansions

Book level

Let $b \geq 2$ and suppose $\sum_{i=0}^r d_i b^i = \sum_{i=0}^r e_i b^i$ with all $d_i, e_i \in \{0, \dots, b-1\}$. Prove that $d_0 = e_0$, and explain how $d_i = e_i$ for all i follows.

Invention: the uniqueness half of the division theorem, the half people forget exists.

Let n be the common value and put $q = \sum_{i=1}^r d_i b^{i-1}$, $q' = \sum_{i=1}^r e_i b^{i-1}$, both in $\mathbb{N}$. Then

$$n = qb + d_0 = q'b + e_0, \quad 0 \leq d_0 < b, \quad 0 \leq e_0 < b.$$

So (q, d_0) and (q', e_0) are both quotient-remainder pairs for n on division by b . By uniqueness in the division theorem, $d_0 = e_0$ and $q = q'$.

The equation $q = q'$ reads $\sum_{i=1}^r d_i b^{i-1} = \sum_{i=1}^r e_i b^{i-1}$: the same kind of equality with one digit fewer. The same argument gives $d_1 = e_1$, then $d_2 = e_2$, and so on. To make “and so on” rigorous, prove by induction on r (Chapter 4) the statement “two expansions with at most $r+1$ digits that have the same value have the same digits”. $\square$

F STRETCH EXERCISES

Attempt these before reading Appendix S. They are pitched at the hardest Chapter 0 exercises.

STRETCH EXERCISES 0 · solutions in Appendix S

- S0.1** Prove that $\log_{10} 2$ is irrational. [Hint: write $2^b = 10^a = 2^a 5^a$ and compare parities after cancelling]
- S0.2** Prove that there is no integer n with $3 \mid n^2 + 1$. [Hint: remainders of n^2 on division by 3]
- S0.3** Let $n \in \mathbb{N}$. Prove that n is even if and only if the sum of its base-3 digits is even. [Hint: 3^i is odd, so $3^i = 2m_i + 1$]
- S0.4** Prove that for all $a, b \in \mathbb{Z}$, if $a \mid b$ and $b \mid a$, then $a = b$ or $a = -b$. [Hint: $a = lka$; treat $a = 0$ separately; then $kl = 1$ in $\mathbb{Z}$]
- S0.5** Prove or disprove: (i) the product of two irrational numbers is irrational; (ii) if x is irrational then x^2 is irrational; (iii) if x is irrational then $1/x$ is irrational.
- S0.6** Prove that for every rational $q > 0$ there exists $n \in \mathbb{N}$ with $n \geq 1$ and $\frac{1}{n} < q$. [Hint: write $q = \frac{a}{b}$ with $a, b \geq 1$ and build n from b]
- S0.7** Prove that there are no integers x, y with $x^2 - y^2 = 2$. [Hint: $x^2 - y^2 = (x-y)(x+y)$, and $x-y, x+y$ have the same parity]

G REVIEW SET 0

- $a \in \{x \in X \mid p(x)\} \Leftrightarrow a \in X \wedge p(a)$.
- $b \in \{e(x) \mid x \in X\} \Leftrightarrow \exists x \in X, b = e(x)$.
- Base b : $n = \sum d_i b^i$, $0 \leq d_i < b$; convert by repeated division; $n = bm + d_0$.
- Division theorem: $a = qb + r$, $0 \leq r < |b|$, unique.
- $b \mid a \Leftrightarrow \exists q \in \mathbb{Z}, a = qb$. Even $= 2k$, odd $= 2k+1$; every integer is exactly one of them.
- Prove $\mid$: exhibit q . Use $\vdash$: fix q .
- Cases by remainder cover all integers.
- Rational $= \frac{a}{b}$, $b \neq 0$; may take lowest terms.
- Irrational: assume $\frac{a}{b}$ in lowest terms, derive contradiction. Lemma: n^2 even $\Rightarrow n$ even (prove the contrapositive).

Chapter 1

Logical structure

Contents

- A** Propositional logic (§1.1)
- B** Variables and quantifiers (§1.2)
- C** Logical equivalence (§1.3)
- D** Book-level examples
- E** Stretch exercises
- F** Review set 1

Planning

Essential: 1.1: 21, 25, 32, 33, 38, 47, 56, 57; 1.3: 5, 13, 14, 22, 27
Essential: 1.2: 6, 15, 18, 39; 1.3: 44ace; 1.E: 18
Recommended: 1.1: 37, 43, 54, 55, 65, 66; 1.2: 21, 22, 28, 37ab; 1.3: 32, 35; 1.E: 11, 28, 29, 30
 §1.1 is in **Quiz 1**.

This chapter is the engine of the course. Every statement you will ever be asked to prove is built from a handful of *connectives* ($\wedge, \vee, \Rightarrow, \Leftrightarrow, \neg$) and *quantifiers* ($\forall, \exists$), and **each of them comes with a fixed proof strategy**. Learn the strategies as a table (two of them below), then every proof starts by reading off the outermost symbol.

A PROPOSITIONAL LOGIC

Propositions and connectives

A **proposition** is a statement that is either true or false. Propositional variables P, Q, R stand for propositions; they are combined with the **logical operators**:

symbol	name	read as	true exactly when
$P \wedge Q$	conjunction	“ P and Q ”	both are true
$P \vee Q$	disjunction	“ P or Q ”	at least one is true (<i>inclusive</i> or)
$P \Rightarrow Q$	implication	“if P then Q ”	P is false, or Q is true
$P \Leftrightarrow Q$	biconditional	“ P if and only if Q ”	P and Q have the same truth value
$\neg P$	negation	“not P ”	P is false

Precedence: $\neg$ binds tightest, then $\wedge$, then $\vee$, then $\Rightarrow$, then $\Leftrightarrow$. So $\neg P \wedge Q \Rightarrow R$ means $((\neg P) \wedge Q) \Rightarrow R$. When in doubt, bracket.

Truth tables

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$	$\neg P$
T	T	T	T	T	T	F
T	F	F	T	F	F	F
F	T	F	T	T	F	T
F	F	F	T	T	T	T

A formula that is true in every row is a **tautology**; false in every row, a **contradiction**. The only row that makes $P \Rightarrow Q$ false is P true, Q false. An implication with a false hypothesis is true (**vacuously true**): “if $1 = 2$ then $\sqrt{2} \in \mathbb{Q}$ ” is a true statement.

Reading implications in English

All of these mean $P \Rightarrow Q$: if P then Q · P implies Q · Q if P · P only if Q · Q whenever P · P is sufficient for Q · Q is necessary for P .

“ P unless Q ” means $\neg Q \Rightarrow P$. The **converse** of $P \Rightarrow Q$ is $Q \Rightarrow P$; it is a different statement.

Proof strategies for the connectives

goal / assumption	to PROVE it	to USE it as an assumption
$P \wedge Q$	prove P ; then prove Q .	you may use both P and Q .
$P \vee Q$	prove P , or prove Q ; usually: <i>assume $\neg P$ and prove Q.</i>	<i>proof by cases</i> : prove the goal assuming P , then prove it again assuming Q .
$P \Rightarrow Q$	“Suppose P .” ... “Hence Q .”	if you also know P , conclude Q (modus ponens). Or use the contrapositive $\neg Q \Rightarrow \neg P$.
$P \Leftrightarrow Q$	prove $P \Rightarrow Q$ and prove $Q \Rightarrow P$ (label the directions).	use either direction.
$\neg P$	“Suppose P .” ... derive a contradiction.	if you also know P : contradiction, and you are done with that case.

Example 1 — truth table

Self Tutor

Construct the truth table of $(P \vee Q) \Rightarrow \neg(P \wedge Q)$. When is the formula false? Is it a tautology?

P	Q	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$(P \vee Q) \Rightarrow \neg(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	T

It is false only when P and Q are both true (“exclusive or” fails there). Not a tautology. □

Example 2 — translating into symbols

Self Tutor

Let $n \in \mathbb{Z}$ and $x \in \mathbb{R}$. Write each statement using the propositions p : “ $4 \mid n$ ”, q : “ n is even”, s : “ $x > 2$ ”, t : “ $x^2 > 4$ ”.

(a) n is even if 4 divides n . (b) 4 divides n only if n is even. (c) $x > 2$ is sufficient for $x^2 > 4$. (d) $x > 2$ is necessary for $x^2 > 4$. Which of (c), (d) is true?

(a) $p \Rightarrow q$. (b) $p \Rightarrow q$ (same statement!). (c) $s \Rightarrow t$: true. (d) $t \Rightarrow s$: false, since $x = -3$ has $x^2 = 9 > 4$ but $x \not> 2$. □

Example 3 — proof by cases (using a disjunction)

Self Tutor

Let $k \in \mathbb{Z}$. Prove that $k(k+1)$ is even.

By the division theorem, k is even or k is odd. We consider the two cases.

Case 1: $k = 2m$ for some $m \in \mathbb{Z}$. Then $k(k+1) = 2m(k+1) = 2(m(k+1))$, which is even.

Case 2: $k = 2m+1$ for some $m \in \mathbb{Z}$. Then $k+1 = 2(m+1)$, so $k(k+1) = 2(k(m+1))$, which is even.

In both cases $k(k+1)$ is even. □

Example 4 — implication and conjunction

Self Tutor

Let $n \in \mathbb{Z}$. Prove that if n is odd, then n^2 is odd and $8 \mid n^2 - 1$.

Suppose n is odd; fix $k \in \mathbb{Z}$ with $n = 2k + 1$. {strategy for $\Rightarrow$: assume the hypothesis}

First conjunct. $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, so n^2 is odd.

Second conjunct. $n^2 - 1 = 4k^2 + 4k = 4k(k + 1)$. By Example 1.3, $k(k + 1) = 2m$ for some $m \in \mathbb{Z}$, so $n^2 - 1 = 8m$ and $8 \mid n^2 - 1$.

Both conjuncts hold, so the conjunction holds. $\square$

The same computation shows: a product of two odd numbers $(2k + 1)(2l + 1) = 2(2kl + k + l) + 1$ is odd.

Example 5 — proving a disjunction

Self Tutor

Let $x \in \mathbb{R}$ and suppose $x^2 = x$. Prove that $x = 0$ or $x = 1$.

We prove the disjunction by assuming $x \neq 0$ and proving $x = 1$. {strategy: $\neg P$, then prove Q }

Suppose $x \neq 0$. From $x^2 = x$ we get $x^2 - x = 0$, so $x(x - 1) = 0$. A product of real numbers is 0 only if a factor is 0; since $x \neq 0$ we must have $x - 1 = 0$, i.e. $x = 1$. $\square$

Example 6 — biconditional

Self Tutor

Let $n \in \mathbb{Z}$. Prove that n is odd if and only if $n + 1$ is even.

$(\Rightarrow)$ Suppose n is odd, say $n = 2k + 1$. Then $n + 1 = 2k + 2 = 2(k + 1)$ is even.

$(\Leftarrow)$ Suppose $n + 1$ is even, say $n + 1 = 2k$. Then $n = 2k - 1 = 2(k - 1) + 1$ with $k - 1 \in \mathbb{Z}$, so n is odd. $\square$

Example 7 — proving a negation

Self Tutor

Prove that no integer is both even and odd.

Suppose, for a contradiction, that $n \in \mathbb{Z}$ is both even and odd: $n = 2k$ and $n = 2l + 1$ for some $k, l \in \mathbb{Z}$. Then $2k = 2l + 1$, so $2(k - l) = 1$, so $k - l = \frac{1}{2}$. But $k - l \in \mathbb{Z}$ and $\frac{1}{2} \notin \mathbb{Z}$: contradiction. Hence no integer is both even and odd. $\square$

Watch out

- To prove $P \Rightarrow Q$ you *assume* P ; you do not prove P , and you do not assume Q . Proving $Q \Rightarrow P$ instead (the converse) earns nothing.
- “Or” is inclusive. To prove $P \vee Q$ you are not obliged to decide which one; the “assume $\neg P$, prove Q ” route is usually easiest.
- Vacuous truth is real: a statement “if P then Q ” about an object for which P fails is true, no proof needed.

EXERCISES 1.1

Essential: 21, 25, 32, 33, 38, 47, 56, 57 **Recommended:** 37, 43, 54, 55, 65, 66 plus **Recommended:** 1.E: 11

Skills: truth tables and tautologies; translating “only if / necessary / sufficient / unless”; writing short proofs whose first line is dictated by the outermost connective ($\Rightarrow$: suppose; $\wedge$: two parts; $\vee$: cases or “assume not P ”; $\Leftrightarrow$: two directions; $\neg$: contradiction).

B VARIABLES AND QUANTIFIERS

Predicates and quantifiers

A **predicate** $p(x)$ is a statement whose truth depends on a variable x from some set X (e.g. $p(x)$: “ $x^2 < 2$ ” on $\mathbb{R}$). A variable is **bound** when a quantifier controls it and **free** otherwise. The quantifiers:

- $\forall x \in X, p(x)$ “for all x in X , $p(x)$ ”: true when $p(a)$ holds for *every* $a \in X$.
- $\exists x \in X, p(x)$ “there exists x in X such that $p(x)$ ”: true when $p(a)$ holds for *at least one* $a \in X$.
- $\exists! x \in X, p(x)$ “there is exactly one”: $\exists x \in X, (p(x) \wedge \forall y \in X, (p(y) \Rightarrow y = x))$.

Over the empty set: $\forall x \in \emptyset, p(x)$ is true (vacuously); $\exists x \in \emptyset, p(x)$ is false.

Translation patterns

English	formula
every P is a Q	$\forall x \in X, (P(x) \Rightarrow Q(x))$
some P is a Q	$\exists x \in X, (P(x) \wedge Q(x))$
no P is a Q	$\forall x \in X, (P(x) \Rightarrow \neg Q(x)) \equiv \neg \exists x \in X, (P(x) \wedge Q(x))$
for every x there is a y (depending on x)	$\forall x, \exists y, \dots$
there is one y that works for every x	$\exists y, \forall x, \dots$

Order matters between $\forall$ and $\exists$: $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$ is true ($y = x + 1$), but $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, y > x$ is false (no largest integer). Two quantifiers of the same kind may be swapped.

Proof strategies for the quantifiers

	to PROVE it	to USE it as an assumption
$\forall x \in X, p(x)$	“Let $x \in X$ (arbitrary).” Prove $p(x)$ using nothing about x except $x \in X$.	plug in any particular $a \in X$ you like: $p(a)$ holds.
$\exists x \in X, p(x)$	exhibit a <i>witness</i> : “Let $x = \dots$. Then $x \in X$ and $p(x)$ because $\dots$ ”.	“Fix $x \in X$ such that $p(x)$.” Give it a name; assume nothing more about it.
$\exists! x \in X, p(x)$	<i>existence</i> (as above) and <i>uniqueness</i> : “Suppose $x, y \in X$ both satisfy p . Then $\dots x = y$.”	you get a witness and the right to say “so $y = x$ ” for any other y with $p(y)$.
disprove $\forall x \in X, p(x)$	give a <i>counterexample</i> : one $a \in X$ with $\neg p(a)$ (justified by $\neg \forall \equiv \exists \neg$, Section C).	

Example 8 — translating quantified statements

Self Tutor

Write in symbols, with $\mathbb{P}$ the set of primes: (a) every even integer greater than 2 is a sum of two primes; (b) there is a real number that is smaller than every positive real number; (c) for every positive real number there is a smaller positive real number. Which of (b), (c) are true?

- (a) $\forall n \in \mathbb{Z}, ((n \text{ even} \wedge n > 2) \Rightarrow \exists p \in \mathbb{P}, \exists q \in \mathbb{P}, n = p + q)$.
 (b) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, (y > 0 \Rightarrow x < y)$: true, witness $x = 0$ (or $x = -1$).
 (c) $\forall y \in \mathbb{R}, (y > 0 \Rightarrow \exists x \in \mathbb{R}, (x > 0 \wedge x < y))$: true, given y take $x = y/2$. □

Example 9 — proving $\forall\exists$ and disproving $\exists\forall$

Self Tutor

(a) Prove: $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, (m > n \wedge m \text{ is even})$. (b) Disprove: $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m > n$.

(a) Let $n \in \mathbb{Z}$. $\{\forall: \text{arbitrary } n\}$ Put $m = 2|n| + 2$. $\{\exists: \text{a witness, allowed to depend on } n\}$ Then $m \in \mathbb{Z}$, m is even, and $m = 2|n| + 2 > |n| \geq n$. So m has both properties. $\square$

(b) We show the negation, $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}, m \leq n$. Let $m \in \mathbb{Z}$ and put $n = m$. Then $n \in \mathbb{Z}$ and $m \leq n$. $\square$

Example 10 — using an $\exists$, proving an $\exists$

Self Tutor

Let $a, b, c \in \mathbb{Z}$. Prove that if $a \mid b$ then $a \mid bc$.

Suppose $a \mid b$. Fix $k \in \mathbb{Z}$ with $b = ka$. $\{\text{using the } \exists \text{ hidden in } a \mid b\}$

Then $bc = (ka)c = (kc)a$, and $kc \in \mathbb{Z}$, so $a \mid bc$. $\{\text{witness } q = kc \text{ for the } \exists \text{ in } a \mid bc\}$ $\square$

Example 11 — unique existence

Self Tutor

Prove that for all $a, b \in \mathbb{R}$ with $a \neq 0$ there is a unique $x \in \mathbb{R}$ such that $ax + b = 0$.

Let $a, b \in \mathbb{R}$ with $a \neq 0$.

Existence. Put $x = -b/a$ (defined since $a \neq 0$). Then $ax + b = -b + b = 0$.

Uniqueness. Suppose $x, y \in \mathbb{R}$ satisfy $ax + b = 0$ and $ay + b = 0$. Subtracting, $a(x - y) = 0$; since $a \neq 0$, $x - y = 0$, so $x = y$. $\square$

Example 12 — counterexample

Self Tutor

Decide whether the statement “for all $n \in \mathbb{N}$, $n^2 + n + 41$ is prime” is true.

It is false. Counterexample: $n = 40$ gives $40^2 + 40 + 41 = 1600 + 81 = 1681 = 41 \cdot 41$, which is not prime. $\{\text{one explicit } n \text{ with } \neg p(n) \text{ suffices}\}$ $\square$

Watch out

- “Let $x \in X$ ” introduces an arbitrary element; from then on you may not choose x , give it extra properties, or “take $x = 3$ ”. If you need a specific element you are proving an $\exists$, not a $\forall$.
- In a $\forall x \exists y$ proof, the witness y may depend on x . In an $\exists y \forall x$ proof it may not: y must be chosen *before* x is introduced.
- Name a witness before using it: “Fix $k \in \mathbb{Z}$ with $b = ka$ ” comes before any line mentioning k . Do not reuse the same letter for two different witnesses.
- A counterexample needs a verification, not just a value: show that p really fails there.

EXERCISES 1.2

Essential: 6, 15, 18, 39 **Recommended:** 21, 22, 28, 37ab **Bonus:** 14 plus **Essential:** 1.E: 18

Skills: identifying free/bound variables; translating English into $\forall/\exists$ formulas (and back) with the right pattern ($\forall \dots \Rightarrow, \exists \dots \wedge$); recognising when quantifier order changes the meaning; writing $\forall$ -proofs with “let”, $\exists$ -proofs with a witness, $\exists!$ -proofs in two parts; disproving with a counterexample.

C LOGICAL EQUIVALENCE

Logical equivalence

Two formulas are **logically equivalent**, written $\varphi \equiv \psi$, if they have the same truth value under every assignment of truth values to their variables (for propositional formulas: identical final columns in the truth table). Equivalent formulas may be substituted for each other inside any statement and inside any proof.

Laws of logical equivalence

For all propositions P, Q, R :

double negation	$\neg\neg P \equiv P$
De Morgan	$\neg(P \wedge Q) \equiv \neg P \vee \neg Q, \quad \neg(P \vee Q) \equiv \neg P \wedge \neg Q$
implication	$P \Rightarrow Q \equiv \neg P \vee Q, \quad \neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
contrapositive	$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$
biconditional	$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$
distributivity	$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R), \quad P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
excluded middle	$P \vee \neg P$ is a tautology; $P \wedge \neg P$ is a contradiction

For predicates $p(x), q(x)$ on a set X :

negating $\forall$	$\neg \forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$
negating $\exists$	$\neg \exists x \in X, p(x) \equiv \forall x \in X, \neg p(x)$
$\forall$ and $\wedge$	$\forall x, (p(x) \wedge q(x)) \equiv (\forall x, p(x)) \wedge (\forall x, q(x))$
$\exists$ and $\vee$	$\exists x, (p(x) \vee q(x)) \equiv (\exists x, p(x)) \vee (\exists x, q(x))$
not laws	$\forall x, (p \vee q) \not\equiv \forall x p \vee \forall x q$ and $\exists x, (p \wedge q) \not\equiv \exists x p \wedge \exists x q$ { counterexample on $\mathbb{Z}$: $p(x)$: x even, $q(x)$: x odd }

The propositional laws are verified by truth tables (Example 1.13). The quantifier laws are the formal version of “not everyone is ...” = “someone is not ...”.

Negating a statement mechanically

Push $\neg$ inwards one symbol at a time: swap $\forall \leftrightarrow \exists$ (keeping the domain), swap $\wedge \leftrightarrow \vee$, replace $\neg(p \Rightarrow q)$ by $p \wedge \neg q$, and finally negate the atomic statements ($\neg(x < y)$ becomes $x \geq y$, $\neg(a = b)$ becomes $a \neq b$). The result has no $\neg$ in front of a quantifier.

Three strategies that come from the laws

- **Contraposition.** To prove $P \Rightarrow Q$ you may instead assume $\neg Q$ and prove $\neg P$. Use it when $\neg Q$ is more concrete than P (“ n is odd” is easier to compute with than “ n^2 is even”).
 - **Contradiction.** To prove P you may assume $\neg P$ and derive a contradiction (R and $\neg R$ for some R). Use it for “not” statements: irrationality, “there is no ...”, “ $X = \emptyset$ ”.
 - **Cases via excluded middle.** For any R , you may split a proof into the case R and the case $\neg R$.
- Always announce the strategy: “We prove the contrapositive.” / “Suppose, for a contradiction, that ...”.

Example 13 — verifying a law with a truth table

Self Tutor

 Show that $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.

P	Q	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$	$\neg Q$	$P \wedge \neg Q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

 The columns for $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$ agree in every row, so the formulas are logically equivalent. $\square$
Example 14 — equivalence by a chain of laws

Self Tutor

 Prove without a truth table that $(P \Rightarrow Q) \wedge (P \Rightarrow R) \equiv P \Rightarrow (Q \wedge R)$.

$$\begin{aligned}
 (P \Rightarrow Q) \wedge (P \Rightarrow R) &\equiv (\neg P \vee Q) \wedge (\neg P \vee R) && \{\text{implication law, twice}\} \\
 &\equiv \neg P \vee (Q \wedge R) && \{\text{distributivity of } \vee \text{ over } \wedge\} \\
 &\equiv P \Rightarrow (Q \wedge R). && \{\text{implication law}\} \quad \square
 \end{aligned}$$

Example 15 — negating a quantified statement

Self Tutor

 Negate: (a) $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, (n \geq N \Rightarrow |a_n - L| < \varepsilon)$; (b) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, (y > x \wedge y^2 < x)$.

(a) $\exists \varepsilon > 0, \forall N \in \mathbb{N}, \exists n \in \mathbb{N}, (n \geq N \wedge |a_n - L| \geq \varepsilon)$. {swap quantifiers; $\neg(p \Rightarrow q) \equiv p \wedge \neg q$; $\neg(<)$ is $\geq$ }
 (b) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, (y \leq x \vee y^2 \geq x)$. {De Morgan on $\wedge$ } $\square$

Example 16 — contraposition

Self Tutor

 Let $n \in \mathbb{Z}$. Prove that if n^2 is even then n is even.

 We prove the contrapositive: if n is odd then n^2 is odd. Suppose n is odd, say $n = 2k + 1$ with $k \in \mathbb{Z}$. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. This proves the contrapositive, hence the statement. $\square$
Example 17 — contradiction

Self Tutor

Prove that there is no smallest positive rational number.

 Suppose, for a contradiction, that q is a smallest positive rational number: $q \in \mathbb{Q}, q > 0$, and $q \leq r$ for every positive $r \in \mathbb{Q}$. Consider $r = q/2$. Then $r \in \mathbb{Q}$ (if $q = \frac{a}{b}$ then $r = \frac{a}{2b}$), $r > 0$, and $r < q$ because $q > 0$. This contradicts $q \leq r$. Hence no smallest positive rational number exists. $\square$
Example 18 — true or false, with proof

Self Tutor

 Is it true that $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y^2 = x$?

 No. The negation is $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y^2 \neq x$, and we prove that. Take $x = -1$. Let $y \in \mathbb{R}$. Then $y^2 \geq 0 > -1$, so $y^2 \neq -1$. $\square$

Watch out

- Contrapositive $\neg Q \Rightarrow \neg P$ is equivalent to $P \Rightarrow Q$. Converse $Q \Rightarrow P$ and inverse $\neg P \Rightarrow \neg Q$ are not.
- Contraposition proves an implication; contradiction proves anything. If you start “suppose not” for an implication, the assumption is $P \wedge \neg Q$ (both parts!), not just $\neg Q$.
- When negating, keep the domain: the negation of $\forall x \in \mathbb{R}, x > 0 \Rightarrow \dots$ is $\exists x \in \mathbb{R}, x > 0 \wedge \dots$, not $\exists x \in \mathbb{R}, x \leq 0 \wedge \dots$.
- A truth table only proves equivalences of *propositional* formulas. Quantifier laws are quoted by name.

EXERCISES 1.3 and 1.E

Essential: 1.3: 5, 13, 14, 22, 27 (up to 1.3.28); 44ace (from 1.3.29); 1.E: 18 **Recommended:** 1.3: 32, 35; 1.E: 11, 28, 29, 30

Skills: proving or refuting $\varphi \equiv \psi$ by truth table or by a chain of laws; using contraposition and contradiction and saying which one you use; negating statements with several quantifiers correctly (44ace-type); deciding “true or false” by proving the statement or its negation.

D BOOK-LEVEL EXAMPLES

The hardest Chapter 1 exercises hide a second connective inside the first (a disjunction inside a contrapositive, three quantifiers deep), or ask you to prove a statement whose truth value you must first decide. Each example names its invention step.

Example 19 — **contrapositive of a disjunction****Book level**

Let $a, b \in \mathbb{Z}$. Prove that if ab is even, then a is even or b is even.

Invention: negate the disjunction with De Morgan before contraposing.

The contrapositive is: if $\neg(a \text{ is even} \vee b \text{ is even})$, then ab is odd. By De Morgan this reads: if a is odd and b is odd, then ab is odd. We prove that. Suppose $a = 2k + 1$ and $b = 2l + 1$ with $k, l \in \mathbb{Z}$. Then

$$ab = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1,$$

which is odd. This proves the contrapositive, hence the statement. □

Example 20 — **a proof that does not tell you the answer****Book level**

Prove that there exist irrational numbers a, b such that a^b is rational.

Invention: split into cases on a statement you cannot decide.

Let $t = \sqrt{2}^{\sqrt{2}}$. By the law of excluded middle, t is rational or t is irrational.

Case 1: t is rational. Take $a = b = \sqrt{2}$, both irrational. Then $a^b = t$ is rational.

Case 2: t is irrational. Take $a = t$ and $b = \sqrt{2}$, both irrational. Then $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, which is rational.

In either case the required a, b exist. □

The proof establishes the $\exists$ without telling us which case holds. Such a proof is called non-constructive.

Example 21 — three quantifiers: prove one, disprove one

Book level

Decide, with proof, whether each statement is true:

 (a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \forall z \in \mathbb{R}, (z > y \Rightarrow z^2 > x)$; (b) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \exists z \in \mathbb{R}, (z > y \wedge z^2 < x)$.

Invention: in (a) the witness y may depend on x but must beat every z ; in (b) prove the negation.

 (a) True. Let $x \in \mathbb{R}$. Put $y = |x| + 1$. Let $z \in \mathbb{R}$ with $z > y$. Then $z > 1$, so $z^2 > z$ (multiply $z > 1$ by $z > 0$); and $z > |x| + 1 > |x| \geq x$. Hence $z^2 > z > x$. $\square$

 (b) False. The negation is $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, \forall z \in \mathbb{R}, (z \leq y \vee z^2 \geq x)$. Let $y \in \mathbb{R}$ and put $x = 0$. Let $z \in \mathbb{R}$. Then $z^2 \geq 0 = x$, so the disjunction holds. $\square$
Example 22 — unique existence with a side condition

Book level

 Let $x \in \mathbb{R}$ with $x \geq 0$. Taking for granted that x has a square root, prove that there is a *unique* $y \in \mathbb{R}$ with $y \geq 0$ and $y^2 = x$.

Invention: for uniqueness, subtract and factorise; then use the side condition $y, y' \geq 0$.

Existence is the granted fact (the book proves it in Chapter 9).

 Uniqueness. Suppose $y, y' \geq 0$ satisfy $y^2 = x = y'^2$. Then $0 = y^2 - y'^2 = (y - y')(y + y')$, so $y - y' = 0$ or $y + y' = 0$. In the first case $y = y'$. In the second, $y + y' = 0$ with $y, y' \geq 0$ forces $y = y' = 0$, so again $y = y'$. $\square$
Example 23 — negating a definition, then using the negation

Book level

 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous at* $a \in \mathbb{R}$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, (|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon).$$

 Write down what it means for f *not* to be continuous at a . Then prove that $H : \mathbb{R} \rightarrow \mathbb{R}$, with $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$, is not continuous at 0.

Invention: after negating, the $\exists \varepsilon$ and $\exists x$ are yours to choose; the $\forall \delta$ is not.

 Not continuous at a : $\exists \varepsilon > 0, \forall \delta > 0, \exists x \in \mathbb{R}, (|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon)$.

 For H at $a = 0$: take $\varepsilon = \frac{1}{2}$. Let $\delta > 0$ be arbitrary. Put $x = -\delta/2$. Then $|x - 0| = \delta/2 < \delta$, while $|H(x) - H(0)| = |0 - 1| = 1 \geq \frac{1}{2}$. So the negation holds, and H is not continuous at 0. $\square$
Example 24 — a law proved from other laws

Book level

 Prove that $(P \wedge Q) \Rightarrow R \equiv P \Rightarrow (Q \Rightarrow R)$, and say what it means for proofs.

$$\begin{aligned} (P \wedge Q) \Rightarrow R &\equiv \neg(P \wedge Q) \vee R && \{\text{implication law}\} \\ &\equiv (\neg P \vee \neg Q) \vee R && \{\text{De Morgan}\} \\ &\equiv \neg P \vee (\neg Q \vee R) && \{\text{associativity of } \vee\} \\ &\equiv \neg P \vee (Q \Rightarrow R) && \{\text{implication law}\} \\ &\equiv P \Rightarrow (Q \Rightarrow R). && \{\text{implication law}\} \quad \square \end{aligned}$$

 For proofs: to prove “if P and Q , then R ” you may assume P and then assume Q , one at a time. And from a hypothesis $P \Rightarrow (Q \Rightarrow R)$ together with P and Q you may conclude R .

Example 25 — infinitely many primes

Book level

Prove that for every $n \in \mathbb{N}$ there is a prime number p with $p > n$.

Invention: the number $n! + 1$.

Let $n \in \mathbb{N}$ and put $N = n! + 1$, where $n! = 1 \cdot 2 \cdots n$ and $0! = 1$. Then $N \geq 2$, so N has a prime factor p (Example 4.7 in Chapter 4 shows every natural number ≥ 2 is a product of primes). We claim $p > n$. Suppose, for a contradiction, that $p \leq n$. Then p is one of the factors $1, 2, \dots, n$ of $n!$, so $p \mid n!$. Also $p \mid N$. Hence $p \mid N - n! = 1$ by Example 0.3, which is impossible since $p \geq 2$. So $p > n$. $\square$

E STRETCH EXERCISES

STRETCH EXERCISES 1 · solutions in Appendix S

- S1.1** Let $a, b \in \mathbb{Z}$. Prove that if $a^2 + b^2$ is odd, then exactly one of a, b is odd. [Hint: prove the contrapositive; “not exactly one” means both even or both odd]
- S1.2** Prove or disprove: (i) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, xy = 1$; (ii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, (x \neq 0 \Rightarrow xy = 1)$; (iii) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, (x \neq 0 \Rightarrow xy = 1)$.
- S1.3** Show that $P \Rightarrow (Q \Rightarrow P)$, $((P \Rightarrow Q) \Rightarrow P) \Rightarrow P$ and $(P \Rightarrow Q) \vee (Q \Rightarrow P)$ are tautologies. [Hint: for each, ask which row could make it false]
- S1.4** Write the negation of each statement and decide which of the two is true: (a) $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, n > x$; (b) $\exists n \in \mathbb{N}, \forall x \in \mathbb{R}, n > x$.
- S1.5** Prove that for all $x, y \in \mathbb{R}$, if $x \neq y$ then there exists $z \in \mathbb{R}$ strictly between x and y . [Hint: cases $x < y$ and $y < x$; use the average]
- S1.6** Let $n \in \mathbb{Z}$. Prove that if $n^2 + 2n$ is odd, then n is odd.
- S1.7** Let $p(x, y)$ be a predicate on a set X . Prove that $(\exists x \in X, \forall y \in X, p(x, y)) \Rightarrow (\forall y \in X, \exists x \in X, p(x, y))$, and give an example showing the converse can fail.

F REVIEW SET 1

- $P \Rightarrow Q$ false only for P true, Q false; “ P only if Q ”, “ Q is necessary for P ” both mean $P \Rightarrow Q$.
- Prove $\wedge$: both. Prove $\vee$: assume $\neg P$, show Q . Prove $\Rightarrow$: suppose P . Prove $\Leftrightarrow$: two directions. Prove $\neg P$: suppose P , contradiction.
- Use $\vee$: cases. Use $\Rightarrow$ with P : get Q .
- Prove $\forall$: “Let $x \in X$ ”. Prove $\exists$: witness. Prove $\exists!$: existence + uniqueness. Use $\exists$: “Fix x with ...”.
- $\forall x \exists y \neq \exists y \forall x$.
- $\neg \forall \equiv \exists \neg$, $\neg \exists \equiv \forall \neg$, $\neg(p \Rightarrow q) \equiv p \wedge \neg q$, De Morgan.
- Contrapositive $\neg Q \Rightarrow \neg P$ (equivalent); converse $Q \Rightarrow P$ (not).
- Counterexample = proof of $\exists x \neg p(x)$.

Chapter 2

Sets

Contents

- A** Sets, subsets and the power set (§2.1)
- B** Set operations (§2.2)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 2

Planning

Essential: 2.1: 10, 27, 31, 35; 2.2: 5, 24, 26, 28, 35, 36, 37, 40, 51; 2.E: 6, 33, 38
Recommended: 2.1: 17, 18, 23, 24, 32, 36; 2.2: 4, 6, 9, 25, 41, 45, 53; 2.E: 4, 5, 16, 28, 32, 41
Bonus: 2.E: 18
 Tested in **Quiz 2** (7 Oct) together with Chapter 3.

Sets are where the logic of Chapter 1 gets exercised: every set statement unfolds to a statement with $\forall$, $\exists$, $\wedge$, $\vee$, $\Rightarrow$, and then the strategies of Chapter 1 take over. The two rules of Section 0A (how to read $\{x \in X \mid p(x)\}$ and $\{e(x) \mid x \in X\}$) are used on almost every line.

A SETS, SUBSETS AND THE POWER SET

Sets, elements, equality

A **set** is a collection of objects, its **elements**; $a \in X$ means a is an element of X . Two sets are equal when they have the same elements (**extensionality**):

$$X = Y \Leftrightarrow \forall a, (a \in X \Leftrightarrow a \in Y).$$

Hence order and repetition do not matter: $\{1, 2, 2, 3\} = \{3, 1, 2\}$. The **empty set** $\emptyset = \{\}$ has no elements. A set is **inhabited** if it has at least one element.

Subsets and the power set

- X is a **subset** of Y , written $X \subseteq Y$, if every element of X is an element of Y : $X \subseteq Y \Leftrightarrow \forall a \in X, a \in Y$.
- $X \subsetneq Y$ (**proper subset**) means $X \subseteq Y$ and $X \neq Y$.
- Facts: $\emptyset \subseteq X$ and $X \subseteq X$ for every set X ; if $X \subseteq Y$ and $Y \subseteq Z$ then $X \subseteq Z$.
- **Equality via double inclusion:** $X = Y \Leftrightarrow X \subseteq Y \wedge Y \subseteq X$.
- The **power set** of X is $\mathcal{P}(X) = \{U \mid U \subseteq X\}$, the set of all subsets of X : $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$.
 $\mathcal{P}(\emptyset) = \{\emptyset\}$, $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$, a set with n elements has 2^n subsets.

Proving things about sets

- $X \subseteq Y$: “Let $a \in X$.” Unfold what $a \in X$ means. ... “So $a \in Y$.” (It is a $\forall$ -proof.)
- $X = Y$: prove $X \subseteq Y$ and $Y \subseteq X$ separately (label them $(\subseteq)$ and $(\supseteq)$); or show $a \in X \Leftrightarrow a \in Y$ by a chain of $\Leftrightarrow$ steps, each justified by a definition or a logical law.
- $X \not\subseteq Y$: exhibit one a with $a \in X$ and $a \notin Y$. $X \neq Y$: exhibit an element of one set that is not in the other.
- $X = \emptyset$: “Suppose $a \in X$ ” and derive a contradiction. X **inhabited**: exhibit an element.
- $X \subsetneq Y$: prove $X \subseteq Y$ and exhibit an element of $Y \setminus X$.

Example 1 — subset and proper subset

Self Tutor

Let $A = \{n \in \mathbb{Z} \mid 6 \mid n\}$ and $B = \{n \in \mathbb{Z} \mid 3 \mid n\}$. Prove that $A \subsetneq B$.

($A \subseteq B$) Let $n \in A$. Then $n \in \mathbb{Z}$ and $6 \mid n$; fix $k \in \mathbb{Z}$ with $n = 6k$. Then $n = 3(2k)$ with $2k \in \mathbb{Z}$, so $3 \mid n$, i.e. $n \in B$.

(proper) $3 \in B$ since $3 = 3 \cdot 1$, but $3 \notin A$ since $3 = 6k$ would force $k = \frac{1}{2} \notin \mathbb{Z}$. So $A \neq B$. $\square$

Example 2 — set equality by double inclusion

Self Tutor

Prove that $\{x \in \mathbb{R} \mid x^2 - 3x + 2 = 0\} = \{1, 2\}$.

($\subseteq$) Let $x \in \mathbb{R}$ with $x^2 - 3x + 2 = 0$. Then $(x-1)(x-2) = 0$, so $x-1 = 0$ or $x-2 = 0$, i.e. $x = 1$ or $x = 2$. Hence $x \in \{1, 2\}$.

($\supseteq$) Let $x \in \{1, 2\}$. If $x = 1$: $1 - 3 + 2 = 0$. If $x = 2$: $4 - 6 + 2 = 0$. In both cases $x \in \mathbb{R}$ and $x^2 - 3x + 2 = 0$. $\square$

Shorter, by equivalences: $x \in \text{LHS} \Leftrightarrow x \in \mathbb{R} \wedge (x-1)(x-2) = 0 \Leftrightarrow x = 1 \vee x = 2 \Leftrightarrow x \in \{1, 2\}$.

Example 3 — power sets and inclusion

Self Tutor

Let X and Y be sets. Prove that if $X \subseteq Y$ then $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Suppose $X \subseteq Y$. Let $U \in \mathcal{P}(X)$; then $U \subseteq X$. {definition of $\mathcal{P}$ } We show $U \subseteq Y$: let $a \in U$. Then $a \in X$ since $U \subseteq X$, and so $a \in Y$ since $X \subseteq Y$. Thus $U \subseteq Y$, i.e. $U \in \mathcal{P}(Y)$. $\square$

Example 4 — $\in$ versus $\subseteq$

Self Tutor

Let $X = \{1, \{2\}\}$. Write down $\mathcal{P}(X)$ and decide: $2 \in X$? $\{2\} \in X$? $\{2\} \subseteq X$? $\emptyset \in \mathcal{P}(X)$? $\emptyset \subseteq \mathcal{P}(X)$?

X has two elements, 1 and $\{2\}$, so $\mathcal{P}(X) = \{\emptyset, \{1\}, \{\{2\}\}, \{1, \{2\}\}\}$.

$2 \in X$: no (2 is not one of the two elements). $\{2\} \in X$: yes. $\{2\} \subseteq X$: no, because $2 \in \{2\}$ but $2 \notin X$. $\emptyset \in \mathcal{P}(X)$: yes, $\emptyset \subseteq X$. $\emptyset \subseteq \mathcal{P}(X)$: yes, $\emptyset$ is a subset of every set. $\square$

Watch out

- $\in$ asks “is it one of the listed things?”; $\subseteq$ asks “is everything in it one of the listed things?”. $\{\emptyset\} \neq \emptyset$: the left set has one element.
- A proof of $X = Y$ with only one inclusion is half a proof. If you use a chain of $\Leftrightarrow$, each step must really be an equivalence, not just $\Rightarrow$.
- “Let $a \in \emptyset$ ” can never happen, so $\emptyset \subseteq X$ holds vacuously. Do not try to pick an element of $\emptyset$.

EXERCISES 2.1

Essential: 10, 27, 31, 35 **Recommended:** 17, 18, 23, 24, 32, 36

Skills: deciding $\in / \subseteq / =$ for concrete (often nested) sets; writing $\subseteq$ -proofs that start with “Let $a \in X$ ”; double-inclusion proofs; listing power sets; proving statements about $\mathcal{P}$ such as Example 2.3.

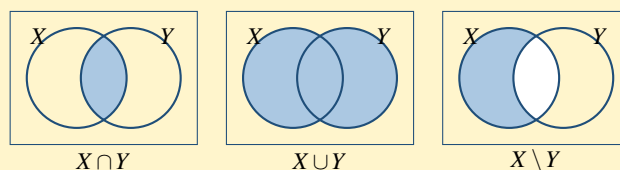
B SET OPERATIONS

Intersection, union, difference

Let X, Y be sets.

operation	definition	membership rule
intersection $X \cap Y$	$\{a \mid a \in X \wedge a \in Y\}$	$a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$
union $X \cup Y$	$\{a \mid a \in X \vee a \in Y\}$	$a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$
relative complement $X \setminus Y$	$\{a \in X \mid a \notin Y\}$	$a \in X \setminus Y \Leftrightarrow a \in X \wedge a \notin Y$

X and Y are **disjoint** if $X \cap Y = \emptyset$.



Algebra of sets

For all sets X, Y, Z :

commutative, associative	$X \cap Y = Y \cap X, X \cup Y = Y \cup X, (X \cap Y) \cap Z = X \cap (Y \cap Z), (X \cup Y) \cup Z = X \cup (Y \cup Z)$
distributive	$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z), X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$
De Morgan	$X \setminus (Y \cup Z) = (X \setminus Y) \cap (X \setminus Z), X \setminus (Y \cap Z) = (X \setminus Y) \cup (X \setminus Z)$
subsets	$X \cap Y \subseteq X \subseteq X \cup Y, X \subseteq Y \Leftrightarrow X \cap Y = X \Leftrightarrow X \cup Y = Y$

Each law is the set version of a logical law ($\cap \leftrightarrow \wedge, \cup \leftrightarrow \vee, \setminus \leftrightarrow \wedge \neg$); that is also how you prove it.

Proving a set identity

Method 1 (membership chain). Take an arbitrary a and rewrite $a \in \text{LHS}$ by definitions until it is a logical formula; transform it with the laws of §1.3; fold it back into $a \in \text{RHS}$. Every step is a $\Leftrightarrow$.

Method 2 (double inclusion). “Let $a \in \text{LHS}$ ” ... “so $a \in \text{RHS}$ ”, then the other way. Membership in a union gives *cases*; membership in an intersection gives *two facts*; membership in $X \setminus Y$ gives a fact and a negated fact.

Use Method 1 when the identity is a direct translation of a logical law; use Method 2 when the two directions need different ideas.

Example 5 — distributive law, both methods

Self Tutor

Prove that $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$ for all sets X, Y, Z .

Method 1. For any a :

$$\begin{aligned}
 a \in X \cap (Y \cup Z) &\Leftrightarrow a \in X \wedge (a \in Y \vee a \in Z) && \{\text{definitions of } \cap, \cup\} \\
 &\Leftrightarrow (a \in X \wedge a \in Y) \vee (a \in X \wedge a \in Z) && \{\text{distributivity, §1.3}\} \\
 &\Leftrightarrow a \in (X \cap Y) \cup (X \cap Z). && \{\text{definitions}\}
 \end{aligned}$$

Since this holds for all a , the sets are equal. □

Method 2. ($\subseteq$) Let $a \in X \cap (Y \cup Z)$. Then $a \in X$ and $a \in Y \cup Z$, so $a \in Y$ or $a \in Z$. If $a \in Y$ then $a \in X \cap Y$;

if $a \in Z$ then $a \in X \cap Z$. Either way $a \in (X \cap Y) \cup (X \cap Z)$.

($\supseteq$) Let $a \in (X \cap Y) \cup (X \cap Z)$. If $a \in X \cap Y$ then $a \in X$ and $a \in Y \subseteq Y \cup Z$. If $a \in X \cap Z$ then $a \in X$ and $a \in Z \subseteq Y \cup Z$. Either way $a \in X$ and $a \in Y \cup Z$, so $a \in X \cap (Y \cup Z)$. $\square$

Example 6 — De Morgan for sets

Self Tutor

Prove that $X \setminus (Y \cup Z) = (X \setminus Y) \cap (X \setminus Z)$.

For any a :

$$\begin{aligned} a \in X \setminus (Y \cup Z) &\Leftrightarrow a \in X \wedge \neg(a \in Y \vee a \in Z) && \{\text{definitions}\} \\ &\Leftrightarrow a \in X \wedge (a \notin Y \wedge a \notin Z) && \{\text{De Morgan}\} \\ &\Leftrightarrow (a \in X \wedge a \notin Y) \wedge (a \in X \wedge a \notin Z) && \{P \wedge Q \wedge R \equiv (P \wedge Q) \wedge (P \wedge R)\} \\ &\Leftrightarrow a \in (X \setminus Y) \cap (X \setminus Z). && \{\text{definitions}\} \quad \square \end{aligned}$$

Example 7 — an “iff” about sets

Self Tutor

Prove that $X \subseteq Y$ if and only if $X \cap Y = X$.

($\Rightarrow$) Suppose $X \subseteq Y$. $X \cap Y \subseteq X$ always holds (if $a \in X \cap Y$ then in particular $a \in X$). For $X \subseteq X \cap Y$: let $a \in X$; then $a \in Y$ by assumption, so $a \in X \cap Y$. Hence $X \cap Y = X$.

($\Leftarrow$) Suppose $X \cap Y = X$. Let $a \in X$. Then $a \in X \cap Y$ by assumption, so $a \in Y$. Hence $X \subseteq Y$. $\square$

Example 8 — refuting a set identity

Self Tutor

Is it true that $X \setminus (Y \setminus Z) = (X \setminus Y) \setminus Z$ for all sets X, Y, Z ?

No. Take $X = Z = \{1\}$ and $Y = \emptyset$. Then $Y \setminus Z = \emptyset$, so $X \setminus (Y \setminus Z) = \{1\}$; but $(X \setminus Y) \setminus Z = \{1\} \setminus \{1\} = \emptyset$. The two sides differ (1 is in the left side only). $\square$

Tip: to find a counterexample, compute both sides as logical formulas; here LHS is $a \in X \wedge (a \notin Y \vee a \in Z)$ and RHS is $a \in X \wedge a \notin Y \wedge a \notin Z$, so put an element in X and Z but not Y .

Cartesian product

The **ordered pair** (a, b) satisfies $(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$ (so $(1, 2) \neq (2, 1)$, unlike $\{1, 2\} = \{2, 1\}$). The **Cartesian product** is

$$X \times Y = \{(a, b) \mid a \in X, b \in Y\}, \quad (a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y.$$

$X^2 = X \times X$; $\mathbb{R}^2$ is the plane. If X has m elements and Y has n , then $X \times Y$ has mn elements; $X \times \emptyset = \emptyset$.

Example 9 — product and intersection

Self Tutor

Prove that $X \times (Y \cap Z) = (X \times Y) \cap (X \times Z)$.

Elements of both sides are ordered pairs. For any pair (a, b) :

$$\begin{aligned} (a, b) \in X \times (Y \cap Z) &\Leftrightarrow a \in X \wedge (b \in Y \wedge b \in Z) \\ &\Leftrightarrow (a \in X \wedge b \in Y) \wedge (a \in X \wedge b \in Z) \quad \{\wedge \text{ is associative, commutative, idempotent}\} \\ &\Leftrightarrow (a, b) \in X \times Y \wedge (a, b) \in X \times Z \Leftrightarrow (a, b) \in (X \times Y) \cap (X \times Z). \quad \square \end{aligned}$$

Indexed families

Let I be a set and let X_i be a set for each $i \in I$ (an **indexed family** $\{X_i \mid i \in I\}$). Then

$$\bigcap_{i \in I} X_i = \{a \mid \forall i \in I, a \in X_i\}, \quad \bigcup_{i \in I} X_i = \{a \mid \exists i \in I, a \in X_i\}.$$

Membership rules: $a \in \bigcap_{i \in I} X_i \Leftrightarrow \forall i \in I, a \in X_i$ and $a \in \bigcup_{i \in I} X_i \Leftrightarrow \exists i \in I, a \in X_i$. Typical index sets: $I = \mathbb{N}$ (write $\bigcup_{n \in \mathbb{N}}$ or $\bigcup_{n \geq 1}$), or $I = \{1, \dots, k\}$. De Morgan: $X \setminus \bigcup_{i \in I} Y_i = \bigcap_{i \in I} (X \setminus Y_i)$ and $X \setminus \bigcap_{i \in I} Y_i = \bigcup_{i \in I} (X \setminus Y_i)$ (these are the quantifier laws $\neg \exists \equiv \forall \neg$, $\neg \forall \equiv \exists \neg$).

Example 10 — computing an indexed intersection and union

Self Tutor

For each $n \in \mathbb{N}$ let $A_n = [n, \infty) = \{x \in \mathbb{R} \mid x \geq n\}$. Prove that $\bigcup_{n \in \mathbb{N}} A_n = [0, \infty)$ and $\bigcap_{n \in \mathbb{N}} A_n = \emptyset$.

Union. ($\subseteq$) Let $x \in \bigcup_n A_n$; fix $n \in \mathbb{N}$ with $x \in A_n$. Then $x \geq n \geq 0$, so $x \in [0, \infty)$. ($\supseteq$) Let $x \in [0, \infty)$. Then $x \geq 0$, so $x \in A_0$, so $x \in \bigcup_n A_n$. {witness $n = 0$ }

Intersection. Suppose, for a contradiction, that $x \in \bigcap_n A_n$. Then $x \geq n$ for every $n \in \mathbb{N}$. But there is a natural number $n > x$ (e.g. $n = \lfloor x \rfloor + 1$ if $x \geq 0$, and $n = 0$ if $x < 0$), and for this n we get $x \geq n > x$: contradiction. So the intersection is empty. $\square$

Example 11 — De Morgan for families

Self Tutor

Prove that $X \setminus \bigcup_{i \in I} Y_i = \bigcap_{i \in I} (X \setminus Y_i)$.

For any a :

$$\begin{aligned} a \in X \setminus \bigcup_{i \in I} Y_i &\Leftrightarrow a \in X \wedge \neg(\exists i \in I, a \in Y_i) \quad \{\text{definitions}\} \\ &\Leftrightarrow a \in X \wedge \forall i \in I, a \notin Y_i \quad \{\neg \exists \equiv \forall \neg\} \\ &\Leftrightarrow \forall i \in I, (a \in X \wedge a \notin Y_i) \quad \{a \in X \text{ does not depend on } i; I \text{ is inhabited}\} \\ &\Leftrightarrow \forall i \in I, a \in X \setminus Y_i \Leftrightarrow a \in \bigcap_{i \in I} (X \setminus Y_i). \quad \square \end{aligned}$$

Example 12 — power sets and operations

Self Tutor

Prove that $\mathcal{P}(X \cap Y) = \mathcal{P}(X) \cap \mathcal{P}(Y)$, and show by example that $\mathcal{P}(X \cup Y) = \mathcal{P}(X) \cup \mathcal{P}(Y)$ can fail.

For any set U : $U \in \mathcal{P}(X \cap Y) \Leftrightarrow U \subseteq X \cap Y \Leftrightarrow (\forall a \in U, a \in X \wedge a \in Y) \Leftrightarrow (U \subseteq X) \wedge (U \subseteq Y) \Leftrightarrow U \in \mathcal{P}(X) \cap \mathcal{P}(Y)$, using $\forall x, (p \wedge q) \equiv \forall x p \wedge \forall x q$.

Counterexample for $\cup$: $X = \{1\}, Y = \{2\}$. Then $\{1, 2\} \in \mathcal{P}(X \cup Y)$, but $\{1, 2\} \not\subseteq X$ and $\{1, 2\} \not\subseteq Y$, so $\{1, 2\} \notin \mathcal{P}(X) \cup \mathcal{P}(Y)$. $\square$

Watch out

- From $a \in X \cup Y$ you may *not* conclude $a \in X$. You get a disjunction: do cases.
- You cannot “cancel”: $X \cup Y = X \cup Z$ does not give $Y = Z$ (take $X = \{1\}$, $Y = \{1\}$, $Z = \emptyset$).
- In a chain of $\Leftrightarrow$ steps, a step that silently uses $X \subseteq Y$ or a specific property of the sets is not a general law. Check that every step holds for *all* sets.
- Elements of $X \times Y$ are pairs; elements of $\mathcal{P}(X)$ are sets. Start those proofs with “Let $(a, b) \in \dots$ ” or “Let $U \in \dots$ ”.

EXERCISES 2.2 and 2.E

Essential: 2.2: 5, 24, 26, 28, 35, 36, 37, 40, 51; 2.E: 6, 33, 38 **Recommended:** 2.2: 4, 6, 9, 25, 41, 45, 53; 2.E: 4, 5, 16, 28, 32, 41 **Bonus:** 2.E: 18

Skills: computing $\cap, \cup, \setminus, \times, \mathcal{P}$ for concrete sets; proving identities by membership chains or double inclusion; refuting false identities with a small counterexample; computing and proving $\cap/\cup$ of families indexed by $\mathbb{N}$ (intervals!); De Morgan in both finite and indexed form; translating $\subseteq$ statements into statements about $\cap, \cup$.

C BOOK-LEVEL EXAMPLES

The hardest Chapter 2 exercises are “iff” statements about sets, families indexed by $\mathbb{N}$, and identities you must first decide are true or false. Each example names its invention step.

Example 13 — an empty product

Book level

Prove that $X \times Y = \emptyset$ if and only if $X = \emptyset$ or $Y = \emptyset$.

Invention: prove $\Rightarrow$ by contraposition, so that the disjunction becomes a conjunction you can use.

($\Leftarrow$) Suppose $X = \emptyset$ or $Y = \emptyset$. Suppose, for a contradiction, that $(a, b) \in X \times Y$. Then $a \in X$ and $b \in Y$, so X and Y are both inhabited, contradicting the assumption. Hence $X \times Y = \emptyset$.

($\Rightarrow$) We prove the contrapositive: if $X \neq \emptyset$ and $Y \neq \emptyset$, then $X \times Y \neq \emptyset$. Fix $a \in X$ and $b \in Y$. Then $(a, b) \in X \times Y$. $\square$

Example 14 — power sets detect inclusion

Book level

Prove that $X \subseteq Y$ if and only if $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Invention: X is itself an element of $\mathcal{P}(X)$.

($\Rightarrow$) is Example 2.3. ($\Leftarrow$) Suppose $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$. Since $X \subseteq X$, we have $X \in \mathcal{P}(X)$, hence $X \in \mathcal{P}(Y)$, which means $X \subseteq Y$. $\square$

Example 15 — intersections and unions of intervals

Book level

Prove that $\bigcap_{n \geq 1} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$ and $\bigcup_{n \geq 1} \left[\frac{1}{n}, 1\right] = (0, 1]$.

Invention: given $x \neq 0$, choose n larger than $1/|x|$; there is a natural number larger than any given real number.

Intersection. ($\supseteq$) For every $n \geq 1$, $-\frac{1}{n} < 0 < \frac{1}{n}$, so 0 lies in every set. ($\subseteq$) Let x lie in the intersection and suppose, for a contradiction, that $x \neq 0$. Then $|x| > 0$; choose $n \in \mathbb{N}$ with $n > 1/|x|$. Then $\frac{1}{n} < |x|$, so $x \notin \left(-\frac{1}{n}, \frac{1}{n}\right)$, contradicting the choice of x . Hence $x = 0$.

Union. ($\subseteq$) If $x \in [\frac{1}{n}, 1]$ for some $n \geq 1$, then $0 < \frac{1}{n} \leq x \leq 1$, so $x \in (0, 1]$. ($\supseteq$) Let $x \in (0, 1]$. Choose $n \geq 1$ with $n \geq 1/x$, i.e. $\frac{1}{n} \leq x$. Then $x \in [\frac{1}{n}, 1]$. $\square$

Example 16 — symmetric difference

Book level

Define $X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$. Prove that $X \triangle Y = (X \cup Y) \setminus (X \cap Y)$.

Invention: both sides say “ a is in exactly one of X, Y ”; check that with a truth table.

Write P for $a \in X$ and Q for $a \in Y$. By the membership rules,

$$a \in X \triangle Y \Leftrightarrow (P \wedge \neg Q) \vee (Q \wedge \neg P), \quad a \in (X \cup Y) \setminus (X \cap Y) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q),$$

and the truth table

P	Q	$(P \wedge \neg Q) \vee (Q \wedge \neg P)$	$(P \vee Q) \wedge \neg(P \wedge Q)$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	F	F

shows the two formulas are logically equivalent. Hence the two sets have the same elements. $\square$

Example 17 — no set contains every set

Book level

Prove that for every set X there is a set R with $R \notin X$.

Invention: Russell’s set $R = \{x \in X \mid x \notin x\}$.

Let X be a set and put $R = \{x \in X \mid x \notin x\}$. Suppose, for a contradiction, that $R \in X$. By the reading rule for set-builder notation, for every x : $x \in R \Leftrightarrow (x \in X \wedge x \notin x)$. Apply this with $x = R$; since $R \in X$, it becomes $R \in R \Leftrightarrow R \notin R$. If $R \in R$ then $R \notin R$, a contradiction; if $R \notin R$ then $R \in R$, a contradiction. Either way we have a contradiction, so $R \notin X$. $\square$

Example 18 — intersection distributes over a union of a family

Book level

Let X be a set and $\{Y_i \mid i \in I\}$ a family of sets. Prove that $X \cap \bigcup_{i \in I} Y_i = \bigcup_{i \in I} (X \cap Y_i)$.

For any a :

$$\begin{aligned} a \in X \cap \bigcup_{i \in I} Y_i &\Leftrightarrow a \in X \wedge \exists i \in I, a \in Y_i \quad \{\text{definitions}\} \\ &\Leftrightarrow \exists i \in I, (a \in X \wedge a \in Y_i) \quad \{P \wedge \exists i Q(i) \equiv \exists i (P \wedge Q(i)) \text{ when } P \text{ does not involve } i\} \\ &\Leftrightarrow \exists i \in I, a \in X \cap Y_i \Leftrightarrow a \in \bigcup_{i \in I} (X \cap Y_i). \quad \square \end{aligned}$$

Example 19 — when the power set of a union splits

Book level

Prove that $\mathcal{P}(X \cup Y) = \mathcal{P}(X) \cup \mathcal{P}(Y)$ if and only if $X \subseteq Y$ or $Y \subseteq X$.

Invention: in the hard direction, build a two-element subset that straddles X and Y .

($\Leftarrow$) If $X \subseteq Y$ then $X \cup Y = Y$ and $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$ (Example 2.3), so both sides equal $\mathcal{P}(Y)$. The case $Y \subseteq X$ is the same with the roles swapped.

($\Rightarrow$) We prove the contrapositive. Suppose $X \not\subseteq Y$ and $Y \not\subseteq X$; fix $a \in X \setminus Y$ and $b \in Y \setminus X$. Then $\{a, b\} \subseteq X \cup Y$, so $\{a, b\} \in \mathcal{P}(X \cup Y)$. But $b \notin X$ gives $\{a, b\} \not\subseteq X$, and $a \notin Y$ gives $\{a, b\} \not\subseteq Y$; so

$\{a, b\} \notin \mathcal{P}(X) \cup \mathcal{P}(Y)$. Hence $\mathcal{P}(X \cup Y) \neq \mathcal{P}(X) \cup \mathcal{P}(Y)$. □

D STRETCH EXERCISES

STRETCH EXERCISES 2 · solutions in Appendix S

S2.1 Prove that $(X \cup Y) \times Z = (X \times Z) \cup (Y \times Z)$.

S2.2 Prove that $X \subseteq Y$ if and only if $X \setminus Y = \emptyset$.

S2.3 Prove that $X \setminus (Y \setminus Z) = (X \setminus Y) \cup (X \cap Z)$.

S2.4 Prove that the sets $[n, n+1)$, $n \in \mathbb{N}$, are pairwise disjoint and that their union is $[0, \infty)$. [Hint: for the union use $n = \lfloor x \rfloor$]

S2.5 Let I be inhabited. Prove that $\bigcap_{i \in I} \mathcal{P}(X_i) = \mathcal{P}(\bigcap_{i \in I} X_i)$. [Hint: two $\forall$'s may be swapped]

S2.6 Prove that $(X \triangle Y) \triangle Z = X \triangle (Y \triangle Z)$. [Hint: show each side is $\{a \mid a \text{ lies in an odd number of } X, Y, Z\}$]

S2.7 Prove or disprove: (i) $(X \setminus Y) \cup Y = X$ for all sets X, Y ; (ii) $(X \cup Y) \setminus Y = X$ if and only if $X \cap Y = \emptyset$.

E REVIEW SET 2

- $X = Y \Leftrightarrow \forall a, (a \in X \Leftrightarrow a \in Y)$; prove by double inclusion or a $\Leftrightarrow$ chain.
- $X \subseteq Y$: “Let $a \in X \dots$ so $a \in Y$ ”.
- $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$; $|\mathcal{P}(X)| = 2^{|X|}$; $\emptyset \in \mathcal{P}(X)$ and $\emptyset \subseteq \mathcal{P}(X)$.
- $\cap \Leftrightarrow \wedge, \cup \Leftrightarrow \vee, \setminus \Leftrightarrow \wedge \neg, \bigcap \Leftrightarrow \forall, \bigcup \Leftrightarrow \exists$.
- Union membership gives cases; intersection gives two facts.
- $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$; pairs are ordered.
- Set laws = logic laws; De Morgan for families = quantifier De Morgan.
- Counterexamples to identities: tiny sets such as $\emptyset, \{1\}, \{2\}$.

Chapter 3

Functions

Contents

- A** Functions, images and preimages (§3.1)
- B** Injections, surjections and bijections (§3.2)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 3

Planning

Essential: 3.1: 11, 15abc, 36; 3.E: 13, 17, 19, 34, 41
Essential: 3.2: 5, 8, 12, 14, 20, 40, 47; 3.E: 24, 25, 48
Recommended: 3.1: 8, 16, 19, 33, 40, 41; 3.2: 6, 16, 19, 25, 39ab, 45, 46; 3.E: 2, 35, 37, 38, 47
Bonus: 3.1: 21
 Tested in **Quiz 2** (7 Oct).

A FUNCTIONS, IMAGES AND PREIMAGES

Function

A **function** $f : X \rightarrow Y$ consists of a set X (the **domain**), a set Y (the **codomain**) and a rule that assigns to each $a \in X$ exactly one element $f(a) \in Y$ (the **value** of f at a). We write $a \mapsto f(a)$ for the rule.

A proposed rule is **well-defined** if (i) it produces a value for every $a \in X$, (ii) the value is unique (does not depend on how a is written or on choices made), and (iii) the value lies in Y .

Two functions are **equal** if they have the same domain, the same codomain and $f(a) = g(a)$ for all $a \in X$. So $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$ and $g : \mathbb{R} \rightarrow [0, \infty), x \mapsto x^2$ are different functions.

Functions you will meet constantly

identity $\text{id}_X : X \rightarrow X, a \mapsto a$; constant functions; the inclusion $U \hookrightarrow X$ for $U \subseteq X$; projections $\pi_1 : X \times Y \rightarrow X, (a, b) \mapsto a$; the **characteristic function** $\chi_U : X \rightarrow \{0, 1\}$ of $U \subseteq X$ ($\chi_U(a) = 1$ if $a \in U$, else 0); the floor $\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$; piecewise rules (“ $f(n) = n/2$ if n even, $f(n) = 3n + 1$ if n odd”); sequences (a_n) as functions $\mathbb{N} \rightarrow \mathbb{R}$; binary operations such as $+: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$.

Example 1 — well-defined or not?

Self Tutor

Which of the following are well-defined functions? (a) $f : \mathbb{Q} \rightarrow \mathbb{Z}, f(\frac{a}{b}) = a$. (b) $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = 1/x$. (c) $h : \mathbb{Z} \rightarrow \mathbb{N}, h(n) = n^2$. (d) $k : \mathbb{R} \rightarrow \mathbb{Z}, k(x) = \text{the integer } n \text{ with } n \leq x < n + 1$.

(a) Not well-defined: $\frac{1}{2} = \frac{2}{4}$ but the rule gives 1 and 2: the value depends on how the input is written (fails (ii)).

(b) Not well-defined: no value at $x = 0$ (fails (i)). It is a function $\mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$.

(c) Well-defined: n^2 is a unique natural number for every integer n (iii: $n^2 \geq 0$).

(d) Well-defined: for each x such an n exists and is unique (if $n \leq x < n + 1$ and $m \leq x < m + 1$ then $|n - m| < 1$, so $n = m$). This is $\lfloor x \rfloor$. □

Graph, composition

- The **graph** of $f : X \rightarrow Y$ is $\text{Gr}(f) = \{(a, f(a)) \mid a \in X\} \subseteq X \times Y$. A subset $G \subseteq X \times Y$ is the graph of a function iff $\forall a \in X, \exists! b \in Y, (a, b) \in G$ (“vertical line test”).
- For $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ the **composite** is $g \circ f : X \rightarrow Z$, $(g \circ f)(a) = g(f(a))$. Composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, and $f \circ \text{id}_X = f = \text{id}_Y \circ f$; it is *not* commutative.

Example 2 — composition is not commutative

Self Tutor

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x + 1$ and $g(x) = x^2$. Compute $g \circ f$ and $f \circ g$ and show $g \circ f \neq f \circ g$.

$(g \circ f)(x) = g(x + 1) = (x + 1)^2$ and $(f \circ g)(x) = f(x^2) = x^2 + 1$. At $x = 1$: $(g \circ f)(1) = 4 \neq 2 = (f \circ g)(1)$. Same domain and codomain, but the values differ at one point, so the functions are different. $\square$

Image and preimage

Let $f : X \rightarrow Y$, $U \subseteq X$ and $V \subseteq Y$.

- The **image** of U is $f[U] = \{f(a) \mid a \in U\} \subseteq Y$. Rule: $b \in f[U] \Leftrightarrow \exists a \in U, f(a) = b$.
- The **preimage** of V is $f^{-1}[V] = \{a \in X \mid f(a) \in V\} \subseteq X$. Rule: $a \in f^{-1}[V] \Leftrightarrow f(a) \in V$.
- $f[X]$ is called *the image of f* . Preimages exist for every function: $f^{-1}[V]$ does not require an inverse function.

Example 3 — computing images and preimages

Self Tutor

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Find $f[[-1, 2]]$, $f^{-1}[[1, 4]]$ and $f^{-1}[\{-1\}]$, with justification for the first.

$f[[-1, 2]] = [0, 4]$. ($\subseteq$) If $x \in [-1, 2]$ then $0 \leq x^2 \leq 4$. ($\supseteq$) If $y \in [0, 4]$ then $\sqrt{y} \in [0, 2] \subseteq [-1, 2]$ and $f(\sqrt{y}) = y$. {witness for the $\exists$ in the image rule}
 $f^{-1}[[1, 4]] = \{x \in \mathbb{R} \mid 1 \leq x^2 \leq 4\} = [-2, -1] \cup [1, 2]$. $f^{-1}[\{-1\}] = \{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$. $\square$

Images and preimages versus set operations

Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $U, U' \subseteq X$ and $V, V' \subseteq Y$. Then:

- | | |
|--|--|
| (a) $U \subseteq U' \Rightarrow f[U] \subseteq f[U']$ | (b) $V \subseteq V' \Rightarrow f^{-1}[V] \subseteq f^{-1}[V']$ |
| (c) $f[U \cup U'] = f[U] \cup f[U']$ | (d) $f^{-1}[V \cup V'] = f^{-1}[V] \cup f^{-1}[V']$ |
| (e) $f[U \cap U'] \subseteq f[U] \cap f[U']$ (equality can fail) | (f) $f^{-1}[V \cap V'] = f^{-1}[V] \cap f^{-1}[V']$ |
| (g) $U \subseteq f^{-1}[f[U]]$ (equality can fail) | (h) $f[f^{-1}[V]] \subseteq V$ (equality can fail) |
| (i) $f^{-1}[Y \setminus V] = X \setminus f^{-1}[V]$ | (j) $(g \circ f)[U] = g[f[U]]$, $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$ |

Preimages behave perfectly; images lose information. Equality in (e) and (g) holds for all U, U' iff f is injective; equality in (h) for all V iff f is surjective (Section B).

Example 4 — an image inclusion and why it is not an equality

Self Tutor

Prove that $f[U \cap U'] \subseteq f[U] \cap f[U']$, and give an example where the inclusion is strict.

Let $b \in f[U \cap U']$. Fix $a \in U \cap U'$ with $f(a) = b$. {image rule} Since $a \in U$ and $f(a) = b$, $b \in f[U]$; since $a \in U'$ and $f(a) = b$, $b \in f[U']$. Hence $b \in f[U] \cap f[U']$. $\square$

Strict: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{-1\}$, $U' = \{1\}$. Then $f[U \cap U'] = f[\emptyset] = \emptyset$ but $f[U] \cap f[U'] = \{1\}$. The proof of “ $\supseteq$ ” breaks because the $a \in U$ and the $a' \in U'$ with $f(a) = f(a') = b$ need not be the same element.

Example 5 — preimage of a complement

Self Tutor

Prove that $f^{-1}[Y \setminus V] = X \setminus f^{-1}[V]$.

For any a : $a \in f^{-1}[Y \setminus V] \Leftrightarrow a \in X \wedge f(a) \in Y \setminus V \Leftrightarrow a \in X \wedge f(a) \in Y \wedge f(a) \notin V \Leftrightarrow a \in X \wedge \neg(f(a) \in V) \Leftrightarrow a \in X \wedge a \notin f^{-1}[V] \Leftrightarrow a \in X \setminus f^{-1}[V]$, using that $f(a) \in Y$ is automatic for $a \in X$. $\square$

Example 6 — $U \subseteq f^{-1}[f[U]]$

Self Tutor

Prove that $U \subseteq f^{-1}[f[U]]$ for every $U \subseteq X$, and find f, U with $U \neq f^{-1}[f[U]]$.

Let $a \in U$. Then $f(a) \in f[U]$ (witness a itself), so $a \in f^{-1}[f[U]]$ by the preimage rule. $\square$

Strict: $f(x) = x^2$ on $\mathbb{R}$, $U = \{1\}$: $f[U] = \{1\}$ and $f^{-1}[\{1\}] = \{-1, 1\} \neq U$.

Watch out

- $f[U]$ lives in the codomain, $f^{-1}[V]$ in the domain. Start a proof about $f[U]$ with “Let $b \in f[U]$; fix $a \in U$ with $f(a) = b$ ”. Start a proof about $f^{-1}[V]$ with “Let $a \in f^{-1}[V]$; then $f(a) \in V$ ”.
- Do not write $f^{-1}(b)$ for a preimage unless f is a bijection. The book’s $f^{-1}[V]$ is a set.
- A function is domain + codomain + rule. Changing the codomain changes the function (and can change surjectivity).

EXERCISES 3.1 and 3.E (first part)

Essential: 3.1: 11, 15abc, 36; 3.E: 13, 17, 19, 34, 41 **Recommended:** 3.1: 8, 16, 19, 33, 40, 41; 3.E: 2, 37, 38 **Bonus:** 3.1: 21

Skills: checking well-definedness; computing composites, images and preimages of concrete sets (intervals, finite sets); proving the inclusions and equalities in the table above from the two membership rules; building counterexamples with $x \mapsto x^2$ or small finite functions.

B INJECTIONS, SURJECTIONS AND BIJECTIONS**Injective, surjective, bijective**

Let $f: X \rightarrow Y$.

- f is **injective** (one-to-one) if $\forall a, b \in X, (f(a) = f(b) \Rightarrow a = b)$ (equivalently $a \neq b \Rightarrow f(a) \neq f(b)$).
Not injective: $\exists a, b \in X, a \neq b \wedge f(a) = f(b)$.
- f is **surjective** (onto) if $\forall b \in Y, \exists a \in X, f(a) = b$ (equivalently $f[X] = Y$).
Not surjective: $\exists b \in Y, \forall a \in X, f(a) \neq b$.
- f is **bijective** if it is both injective and surjective.

Templates

- **Injective:** “Let $a, b \in X$ and suppose $f(a) = f(b)$.” Compute until “ $a = b$ ”. (Do not start from $a = b$.)
- **Not injective:** two explicit different inputs with the same output.
- **Surjective:** “Let $b \in Y$.” Solve $f(a) = b$ for a on scratch paper; in the proof: “Put $a = \dots$. Then $a \in X$ {check!} and $f(a) = \dots = b$.”
- **Not surjective:** one explicit $b \in Y$ and a proof that $f(a) \neq b$ for all a (often a parity or sign argument).
- **Bijjective:** both of the above, *or* exhibit an inverse (see below) and verify both compositions.

Example 7 — a bijection, directly

Self Tutor

Prove that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x - 2$ is bijective.

Injective. Let $x, y \in \mathbb{R}$ with $f(x) = f(y)$. Then $3x - 2 = 3y - 2$, so $3x = 3y$, so $x = y$.

Surjective. Let $y \in \mathbb{R}$. Put $x = \frac{y+2}{3} \in \mathbb{R}$. Then $f(x) = 3 \cdot \frac{y+2}{3} - 2 = y$.

Hence f is bijective. □

Example 8 — neither injective nor surjective

Self Tutor

Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, $g(n) = n^2 + n$. Show that g is neither injective nor surjective.

Not injective: $g(0) = 0 = g(-1)$ and $0 \neq -1$.

Not surjective: we show $1 \notin g[\mathbb{Z}]$. Let $n \in \mathbb{Z}$. Then $g(n) = n(n+1)$ is even by Example 1.3, while 1 is odd, so $g(n) \neq 1$. □

Example 9 — a bijection $\mathbb{Z} \rightarrow \mathbb{N}$ with cases

Self Tutor

Define $h : \mathbb{Z} \rightarrow \mathbb{N}$ by $h(n) = 2n$ if $n \geq 0$ and $h(n) = -2n - 1$ if $n < 0$. Prove that h is a bijection.

First, h is well-defined: for $n \geq 0$, $2n \in \mathbb{N}$; for $n < 0$, $-2n - 1 \geq 2 - 1 = 1 \in \mathbb{N}$. Note $h(n)$ is even when $n \geq 0$ and odd when $n < 0$.

Injective. Let $m, n \in \mathbb{Z}$ with $h(m) = h(n)$. Since $h(m) = h(n)$ have the same parity, m, n are both ≥ 0 or both < 0 . If both ≥ 0 : $2m = 2n$, so $m = n$. If both < 0 : $-2m - 1 = -2n - 1$, so $m = n$.

Surjective. Let $k \in \mathbb{N}$. If k is even, $k = 2j$ with $j \in \mathbb{N}$, and $h(j) = 2j = k$. If k is odd, $k = 2j + 1$ with $j \in \mathbb{N}$; put $n = -(j + 1) < 0$, then $h(n) = 2(j + 1) - 1 = k$. □

Inverses

Let $f : X \rightarrow Y$. A function $g : Y \rightarrow X$ is

- a **left inverse** of f if $g \circ f = \text{id}_X$ (i.e. $g(f(a)) = a$ for all $a \in X$);
- a **right inverse** of f if $f \circ g = \text{id}_Y$ (i.e. $f(g(b)) = b$ for all $b \in Y$);
- an **inverse** of f if it is both. Then f has exactly one inverse, written f^{-1} , and $(f^{-1})^{-1} = f$.

Inverses and the three properties

Let $f : X \rightarrow Y$ with X inhabited.

- 1 f is injective $\Leftrightarrow f$ has a left inverse.
- 2 f is surjective $\Leftrightarrow f$ has a right inverse.
- 3 f is bijective $\Leftrightarrow f$ has an inverse. Then f^{-1} is also a bijection and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ for bijections f, g .

Composition

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.

- 1 If f and g are injective (surjective, bijective), then so is $g \circ f$.
- 2 If $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.

Example 10 — injectivity of a composite

Self Tutor

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Prove that if $g \circ f$ is injective then f is injective. Show by example that g need not be injective.

Suppose $g \circ f$ is injective. Let $a, b \in X$ with $f(a) = f(b)$. Applying g gives $g(f(a)) = g(f(b))$, i.e. $(g \circ f)(a) = (g \circ f)(b)$, so $a = b$ by injectivity of $g \circ f$. Hence f is injective. $\square$

Example: $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$, $f(0) = 0$, $g(0) = g(1) = 0$. Then $g \circ f$ is injective (one-element domain) but g is not.

Example 11 — left inverse $\Rightarrow$ injective

Self Tutor

Let $f : X \rightarrow Y$ and suppose $g : Y \rightarrow X$ satisfies $g \circ f = \text{id}_X$. Prove that f is injective.

Let $a, b \in X$ with $f(a) = f(b)$. Then $a = \text{id}_X(a) = g(f(a)) = g(f(b)) = \text{id}_X(b) = b$. $\square$

Converse idea: if f is injective and $x_0 \in X$, define $g(b) =$ the unique a with $f(a) = b$ when $b \in f[X]$, and $g(b) = x_0$ otherwise; then $g \circ f = \text{id}_X$.

Example 12 — bijective via an explicit inverse

Self Tutor

Let $f : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{1\}$, $f(x) = \frac{x}{x-2}$. Prove that f is a bijection and find f^{-1} .

Scratch: solve $y = \frac{x}{x-2}$: $y(x-2) = x \Rightarrow x(y-1) = 2y \Rightarrow x = \frac{2y}{y-1}$.

Define $g : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R} \setminus \{2\}$ by $g(y) = \frac{2y}{y-1}$. It is well-defined: $y \neq 1$, and $g(y) = 2$ would mean $2y = 2y - 2$, impossible, so $g(y) \neq 2$.

For $x \neq 2$: $g(f(x)) = \frac{2 \cdot \frac{x}{x-2}}{\frac{x}{x-2} - 1} = \frac{2x}{x - (x-2)} = x$. For $y \neq 1$: $f(g(y)) = \frac{\frac{2y}{y-1}}{\frac{2y}{y-1} - 2} = \frac{2y}{2y - 2(y-1)} = y$.

So $g \circ f = \text{id}$ and $f \circ g = \text{id}$; hence f is a bijection with $f^{-1} = g$. $\square$

Example 13 — injective $\Rightarrow$ images preserve intersections

Self Tutor

Let $f : X \rightarrow Y$ be injective. Prove that $f[U \cap U'] = f[U] \cap f[U']$ for all $U, U' \subseteq X$.

“ $\subseteq$ ” is Example 3.4. ($\supseteq$) Let $b \in f[U] \cap f[U']$. Fix $a \in U$ with $f(a) = b$ and $a' \in U'$ with $f(a') = b$. Then $f(a) = f(a')$, so $a = a'$ by injectivity. Hence $a \in U \cap U'$ and $f(a) = b$, so $b \in f[U \cap U']$. $\square$

Watch out

- In an injectivity proof the hypothesis is $f(a) = f(b)$ and the goal is $a = b$. Writing “ $a = b \Rightarrow f(a) = f(b)$ ” proves nothing (every function does that).
- Surjectivity is about the *codomain*: $x \mapsto x^2$ is surjective $\mathbb{R} \rightarrow [0, \infty)$ but not $\mathbb{R} \rightarrow \mathbb{R}$. Always check that your witness a lies in the domain.
- f^{-1} as a function exists only for bijections; “ $f^{-1}(b)$ ” in a proof must be preceded by “since f is bijective”.
- For piecewise functions, injectivity needs the cross-cases too (an input from each piece); separate the pieces by a property of the outputs, as in Example 3.9.

EXERCISES 3.2 and 3.E (second part)

Essential: 3.2: 5, 8, 12, 14, 20, 40, 47; 3.E: 24, 25, 48 **Recommended:** 3.2: 6, 16, 19, 25, 39ab, 45, 46; 3.E: 35, 47

Skills: proving or refuting injectivity/surjectivity for concrete functions on $\mathbb{N}, \mathbb{Z}, \mathbb{R}$, products and power sets; finding inverses and verifying both compositions; proving the general facts about composites and left/right inverses; combining injectivity/surjectivity with images and preimages (equality cases of the Section A table).

C BOOK-LEVEL EXAMPLES

The hardest Chapter 3 exercises characterise injectivity or surjectivity by a property of *all* images, preimages or compositions, or ask for a bijection between two sets that look different. The invention is usually “apply the hypothesis to a cleverly chosen small set”.

Example 14 — surjectivity through preimages

Book level

Let $f : X \rightarrow Y$. Prove that f is surjective if and only if $f[f^{-1}[V]] = V$ for every $V \subseteq Y$.

Invention: in the $\Leftarrow$ direction, apply the hypothesis to $V = \{b\}$.

$(\Rightarrow)$ Suppose f is surjective and let $V \subseteq Y$. The inclusion $f[f^{-1}[V]] \subseteq V$ holds for every f : if $b \in f[f^{-1}[V]]$, fix $a \in f^{-1}[V]$ with $f(a) = b$; then $b = f(a) \in V$. For $\supseteq$: let $b \in V$. By surjectivity fix $a \in X$ with $f(a) = b$. Then $f(a) \in V$, so $a \in f^{-1}[V]$, so $b = f(a) \in f[f^{-1}[V]]$.

$(\Leftarrow)$ Suppose $f[f^{-1}[V]] = V$ for all $V \subseteq Y$, and let $b \in Y$. Apply the hypothesis to $V = \{b\}$: then $b \in f[f^{-1}[\{b\}]]$, so there is $a \in f^{-1}[\{b\}]$ with $f(a) = b$. This a witnesses surjectivity at b . $\square$

Example 15 — injectivity through images

Book level

Let $f : X \rightarrow Y$. Prove that f is injective if and only if $f^{-1}[f[U]] = U$ for every $U \subseteq X$.

Invention: apply the hypothesis to a one-element set.

$(\Rightarrow)$ Suppose f is injective and $U \subseteq X$. $U \subseteq f^{-1}[f[U]]$ holds always (Example 3.6). Conversely let $a \in f^{-1}[f[U]]$, so $f(a) \in f[U]$; fix $u \in U$ with $f(u) = f(a)$. By injectivity $a = u \in U$.

$(\Leftarrow)$ Suppose $f^{-1}[f[U]] = U$ for all U . Let $a, a' \in X$ with $f(a) = f(a')$. Then $f(a') \in f[\{a\}]$, so $a' \in f^{-1}[f[\{a\}]] = \{a\}$, hence $a' = a$. $\square$

Example 16 — a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$

Book level

Define $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by $g(m, n) = 2^m(2n + 1) - 1$. Prove that g is a bijection.

Invention: a lemma about the shape of positive integers, proved by strong induction.

Lemma. Every integer $k \geq 1$ can be written as $k = 2^m(2n + 1)$ with $m, n \in \mathbb{N}$, and m, n are unique. *Existence* is Example 4.16 (Chapter 4). *Uniqueness:* suppose $2^m(2n + 1) = 2^{m'}(2n' + 1)$ and, without loss of generality, $m \leq m'$. Dividing by 2^m gives $2n + 1 = 2^{m'-m}(2n' + 1)$. If $m' > m$, the right-hand side is even and the left-hand side is odd, impossible; so $m' = m$, and then $2n + 1 = 2n' + 1$ gives $n = n'$.

Injective. If $g(m, n) = g(m', n')$ then $2^m(2n + 1) = 2^{m'}(2n' + 1)$, so $(m, n) = (m', n')$ by uniqueness.

Surjective. Let $k \in \mathbb{N}$. Then $k + 1 \geq 1$, so $k + 1 = 2^m(2n + 1)$ for some $m, n \in \mathbb{N}$ by existence, i.e. $g(m, n) = k$. $\square$

Example 17 — subsets are the same as $\{0, 1\}$ -valued functions

Book level

Let X be a set and write $\{0, 1\}^X$ for the set of all functions $X \rightarrow \{0, 1\}$. Prove that $\chi : \mathcal{P}(X) \rightarrow \{0, 1\}^X$, $U \mapsto \chi_U$, is a bijection.

Invention: write down the inverse, $g \mapsto g^{-1}[\{1\}]$, and check both composites.

Define $\psi : \{0, 1\}^X \rightarrow \mathcal{P}(X)$ by $\psi(g) = g^{-1}[\{1\}] = \{a \in X \mid g(a) = 1\}$.

$\psi \circ \chi = \text{id}$: for $U \subseteq X$, $\psi(\chi_U) = \{a \in X \mid \chi_U(a) = 1\} = \{a \in X \mid a \in U\} = U$.

$\chi \circ \psi = \text{id}$: let $g : X \rightarrow \{0, 1\}$ and $a \in X$. If $g(a) = 1$ then $a \in \psi(g)$, so $\chi_{\psi(g)}(a) = 1 = g(a)$. If $g(a) = 0$ then $a \notin \psi(g)$, so $\chi_{\psi(g)}(a) = 0 = g(a)$. Thus $\chi_{\psi(g)} = g$ (same domain, codomain and values).

So χ has an inverse and is a bijection. $\square$

Consequence: a set with n elements has 2^n subsets, because there are 2^n functions from it to $\{0, 1\}$ (two choices per element).

Example 18 — injective means left-cancellable

Book level

Let $f : X \rightarrow Y$. Prove that f is injective if and only if for every set Z and all functions $g, h : Z \rightarrow X$: if $f \circ g = f \circ h$ then $g = h$.

Invention: in the $\Leftarrow$ direction, use a one-point set Z .

$(\Rightarrow)$ Suppose f is injective and $f \circ g = f \circ h$ with $g, h : Z \rightarrow X$. Let $z \in Z$. Then $f(g(z)) = f(h(z))$, so $g(z) = h(z)$. Since z was arbitrary and g, h share domain and codomain, $g = h$.

$(\Leftarrow)$ Suppose the cancellation property holds. Let $a, a' \in X$ with $f(a) = f(a')$. Let $Z = \{*\}$ and define $g, h : Z \rightarrow X$ by $g(*) = a$ and $h(*) = a'$. Then $(f \circ g)(*) = f(a) = f(a') = (f \circ h)(*)$, so $f \circ g = f \circ h$, so $g = h$ by hypothesis, so $a = g(*) = h(*) = a'$. $\square$

Example 19 — one-sided inverses that meet in the middle

Book level

Let $f : X \rightarrow Y$. Suppose $g : Y \rightarrow X$ satisfies $g \circ f = \text{id}_X$ and $h : Y \rightarrow X$ satisfies $f \circ h = \text{id}_Y$. Prove that $g = h$, and deduce that f is a bijection with inverse g .

Invention: insert an identity and re-bracket.

$g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h$, using associativity of composition.

Hence $g \circ f = \text{id}_X$ and $f \circ g = f \circ h = \text{id}_Y$, so g is a two-sided inverse of f ; thus f is a bijection and $f^{-1} = g$. The same computation shows that inverses are unique: if g and g' are both inverses of f , run the argument with $h = g'$. $\square$

D STRETCH EXERCISES**STRETCH EXERCISES 3 · solutions in Appendix S**

S3.1 Let $f : X \rightarrow Y$ and $V \subseteq Y$. Prove that $f[f^{-1}[V]] = V \cap f[X]$.

S3.2 Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. [Hint: check both composites and use uniqueness of inverses]

S3.3 Let $f : X \rightarrow Y$. Prove that f is injective if and only if $f[X \setminus U] \subseteq Y \setminus f[U]$ for every $U \subseteq X$. [Hint: for $\Leftarrow$, take $U = \{a'\}$]

S3.4 Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(n) = n + 1$ if n is even and $f(n) = n - 1$ if n is odd. Prove that $f \circ f = \text{id}_{\mathbb{Z}}$ and deduce that f is a bijection.

S3.5 Let $f : X \rightarrow Y$ have a left inverse g and a right inverse h . Prove that $g = h$.

- S3.6** Prove that $f : X \rightarrow Y$ is surjective if and only if for every set Z and all $g, h : Y \rightarrow Z$: $g \circ f = h \circ f$ implies $g = h$. [Hint: for $\Leftarrow$, suppose $b \notin f[X]$ and build $g, h : Y \rightarrow \{0, 1\}$ that differ only at b]
- S3.7** Prove that for every set X there is no surjection $f : X \rightarrow \mathcal{P}(X)$. [Hint: consider $D = \{a \in X \mid a \notin f(a)\}$]
- S3.8** Let $a < b$ be real numbers. Prove that $x \mapsto a + (b - a)x$ is a bijection $(0, 1) \rightarrow (a, b)$ by exhibiting its inverse.

E**REVIEW SET 3**

- Function = domain + codomain + rule; well-defined = every input, one output, in the codomain.
- $b \in f[U] \Leftrightarrow \exists a \in U, f(a) = b$; $a \in f^{-1}[V] \Leftrightarrow f(a) \in V$.
- Preimages commute with $\cap, \cup, \setminus$; images only with $\cup$ (and $\subseteq$ for $\cap$).
- Injective: $f(a) = f(b) \Rightarrow a = b$. Surjective: every b has an a . Bijective: both $\Leftrightarrow$ inverse exists.
- Left inverse $\Leftrightarrow$ injective; right inverse $\Leftrightarrow$ surjective.
- $g \circ f$ injective $\Rightarrow f$ injective; $g \circ f$ surjective $\Rightarrow g$ surjective.
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- Counterexample factory: $x \mapsto x^2, n \mapsto 2n, n \mapsto n(n+1)$, tiny finite sets.

Chapter 4

Mathematical induction

Contents

- A** Weak induction (§4.2)
- B** Strong induction and well-ordering (§4.3)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 4

Planning

Essential: 4.2: 5, 11; 4.E: 10, 11, 16, 17, 18, 20, 25
Essential: 4.3: 8, 9, 14, 15; 4.E: 13, 19, 26, 27, 29, 30, 31, 32
Recommended: 4.2: 8, 17, 20; 4.E: 12, 14, 15, 21, 24, 33; 4.3: 10, 18; 4.E: 28, 34, 35
 (§4.1, Peano's axioms, is not in the planning.)

A WEAK INDUCTION

Principle of (weak) induction

Let $p(n)$ be a statement about natural numbers n and let $b \in \mathbb{N}$. Suppose

- **(base case)** $p(b)$ is true, and
- **(induction step)** for all $n \geq b$: if $p(n)$ is true then $p(n+1)$ is true.

Then $p(n)$ is true for all $n \geq b$. (Most often $b = 0$ or $b = 1$.)

In the induction step the assumption “ $p(n)$ ” is the **induction hypothesis**. You prove $p(n+1)$ *from* it, for an arbitrary $n \geq b$; you do not get to assume $p(n+1)$.

The induction template (use these words)

- 1 “For $n \in \mathbb{N}$ (with $n \geq b$) let $p(n)$ be the statement: ...”. $p(n)$ is a *statement*, not a number.
- 2 “**Base case** $n = b$: ...” Verify $p(b)$ by direct computation.
- 3 “**Induction step.** Let $n \geq b$ and suppose $p(n)$ holds, i.e. ... (IH). We prove $p(n+1)$, i.e. ...”
Start from the $n+1$ expression, isolate the n expression inside it, apply the IH, finish with algebra.
- 4 “By induction, $p(n)$ holds for all $n \geq b$.” $\square$

Notation that appears in the exercises

$\sum_{k=1}^n a_k = a_1 + \cdots + a_n$ and $\prod_{k=1}^n a_k = a_1 \cdots a_n$; an empty sum is 0, an empty product is 1. Recursive definitions: $0! = 1$, $(n+1)! = (n+1) \cdot n!$; a sequence given by a_0 and a rule for a_{n+1} in terms of a_n . The key identity for the step is $\sum_{k=1}^{n+1} a_k = (\sum_{k=1}^n a_k) + a_{n+1}$.

Example 1 — a sum formula

Self Tutor

Prove that $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.

For $n \in \mathbb{N}$ let $p(n)$ be the statement $\sum_{k=1}^n k = \frac{n(n+1)}{2}$.
 Base case $n = 0$: the empty sum is 0 and $\frac{0 \cdot 1}{2} = 0$. So $p(0)$ holds.

Induction step. Let $n \in \mathbb{N}$ and suppose $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ (IH). Then

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^n k \right) + (n+1) \stackrel{\text{IH}}{=} \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2} = \frac{(n+1)((n+1)+1)}{2},$$

which is $p(n+1)$.

By induction, $p(n)$ holds for all $n \in \mathbb{N}$. □

Example 2 — geometric sum

Self Tutor

Let $r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{k=0}^n r^k = \frac{r^{n+1} - 1}{r - 1}$ for all $n \in \mathbb{N}$.

Base case $n = 0$: LHS = $r^0 = 1$, RHS = $\frac{r-1}{r-1} = 1$.

Induction step: let $n \in \mathbb{N}$ and assume the formula for n . Then $\sum_{k=0}^{n+1} r^k = \frac{r^{n+1} - 1}{r - 1} + r^{n+1} = \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r - 1} = \frac{r^{n+2} - 1}{r - 1}$, the formula for $n + 1$.

By induction the formula holds for all $n \in \mathbb{N}$. □

Example 3 — divisibility

Self Tutor

Prove that $3 \mid 4^n - 1$ for all $n \in \mathbb{N}$.

Base case $n = 0$: $4^0 - 1 = 0 = 3 \cdot 0$.

Induction step: let $n \in \mathbb{N}$ and suppose $3 \mid 4^n - 1$; fix $k \in \mathbb{Z}$ with $4^n - 1 = 3k$. Then

$$4^{n+1} - 1 = 4 \cdot 4^n - 1 = 4(4^n - 1) + 3 = 4 \cdot 3k + 3 = 3(4k + 1),$$

so $3 \mid 4^{n+1} - 1$. By induction, $3 \mid 4^n - 1$ for all $n \in \mathbb{N}$. □

Trick: write the $n + 1$ quantity as (something) · (the n quantity) + (something divisible).

Example 4 — an inequality with a larger base case

Self Tutor

Prove that $2^n > n^2$ for all $n \geq 5$.

Base case $n = 5$: $2^5 = 32 > 25 = 5^2$.

Induction step: let $n \geq 5$ and suppose $2^n > n^2$. Then $2^{n+1} = 2 \cdot 2^n > 2n^2$, so it suffices to show $2n^2 \geq (n+1)^2$, i.e. $n^2 - 2n - 1 \geq 0$, i.e. $(n-1)^2 \geq 2$, which holds since $n-1 \geq 4$. Hence $2^{n+1} > (n+1)^2$.

By induction, $2^n > n^2$ for all $n \geq 5$. □

Pattern for inequalities: bound the $n + 1$ side by the IH, then prove a separate “gap” inequality for $n \geq b$.

Example 5 — closed form of a recursive sequence

Self Tutor

Let $a_0 = 1$ and $a_{n+1} = 2a_n + 1$ for $n \in \mathbb{N}$. Prove that $a_n = 2^{n+1} - 1$ for all $n \in \mathbb{N}$.

Base case: $a_0 = 1 = 2^1 - 1$. Induction step: let $n \in \mathbb{N}$ with $a_n = 2^{n+1} - 1$. Then $a_{n+1} = 2a_n + 1 = 2(2^{n+1} - 1) + 1 = 2^{n+2} - 1$. By induction, $a_n = 2^{n+1} - 1$ for all n . □

Binomial coefficients and the binomial theorem

For $n, k \in \mathbb{N}$ with $k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the number of k -element subsets of an n -element set ($\binom{n}{k} = 0$ for $k > n$).

Pascal's rule: $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$. **Binomial theorem:** for $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Both are proved by induction on n (the theorem's step uses Pascal's rule). Consequences: $\sum_{k=0}^n \binom{n}{k} = 2^n$ ($x = y = 1$), $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \geq 1$.

Example 6 — Bernoulli's inequality

Self Tutor

Let $x \in \mathbb{R}$ with $x \geq -1$. Prove that $(1+x)^n \geq 1+nx$ for all $n \in \mathbb{N}$.

Base case $n = 0$: $(1+x)^0 = 1 \geq 1+0$. Induction step: let $n \in \mathbb{N}$ and suppose $(1+x)^n \geq 1+nx$. Since $1+x \geq 0$ we may multiply the IH by $1+x$ without flipping the inequality:

$$(1+x)^{n+1} = (1+x)(1+x)^n \geq (1+x)(1+nx) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

as $nx^2 \geq 0$. By induction the inequality holds for all $n \in \mathbb{N}$. □

Watch out

- No base case, or the wrong base case ($n = 0$ when the claim is for $n \geq 1$): the proof is invalid. The classic “all horses have the same colour” fallacy has a correct-looking step that fails exactly from $n = 1$ to $n = 2$.
- Never start the step with the statement $p(n+1)$ and manipulate it to something true; start from one side of $p(n+1)$ and transform it, or show both sides equal a common expression.
- State $p(n)$ explicitly and say where you use the IH. “Assume true for n ” without saying what is true is not acceptable.
- If the step uses $p(n-1)$ as well as $p(n)$, you are doing strong induction: see Section B for the extra base cases.

EXERCISES 4.2 and 4.E (weak induction)

Essential: 4.2: 5, 11; 4.E: 10, 11, 16, 17, 18, 20, 25 **Recommended:** 4.2: 8, 17, 20; 4.E: 12, 14, 15, 21, 24, 33

Skills: sum/product formulas, divisibility, inequalities with a base case $b > 0$, recursively defined sequences, statements about finite sets (e.g. number of subsets), binomial coefficients and the binomial theorem. Every one of them is the template above with a different algebraic middle.

B STRONG INDUCTION AND WELL-ORDERING**Principle of strong induction**

Let $p(n)$ be a statement about natural numbers and $b \in \mathbb{N}$. Suppose that for every $n \geq b$:

if $p(k)$ holds for all k with $b \leq k < n$, then $p(n)$ holds.

Then $p(n)$ holds for all $n \geq b$.

In practice: prove the first few values $p(b), \dots, p(b+m)$ directly (**base cases**), then for $n > b+m$ prove $p(n)$ assuming $p(k)$ for all $b \leq k < n$ (or just for the specific smaller k you need, such as $n-1$ and $n-2$).

Weak and strong induction are equivalent principles, but strong induction is the right tool whenever the “reason” for $p(n)$ involves a smaller value that is not $n-1$: factoring $n = ab$, the recurrence $f_n = f_{n-1} + f_{n-2}$, subtracting 4 from n , dividing n by b .

Strong induction template

- 1 “Let $p(n)$ be the statement ...”. Decide which smaller values the step will need, and hence how many base cases.
- 2 “**Base cases.**” Check each one directly.
- 3 “**Induction step.** Let $n \geq \dots$ and suppose $p(k)$ holds for all $b \leq k < n$ (IH). ... Therefore $p(n)$.” Say explicitly which $k < n$ you apply the IH to, and check that $k \geq b$.
- 4 “By strong induction, $p(n)$ holds for all $n \geq b$.”

Example 7 — prime factorisation exists

Self Tutor

Prove that every natural number $n \geq 2$ is a product of (one or more) primes.

Let $p(n)$: “ n is a product of primes”. Let $n \geq 2$ and suppose $p(k)$ for all $2 \leq k < n$ (IH).

Case 1: n is prime. Then n is a product of one prime.

Case 2: n is not prime. Then $n = ab$ with $1 < a, b < n$ (a positive divisor other than 1 and n exists). By the IH, a and b are products of primes, so $n = ab$ is too.

In both cases $p(n)$ holds, so by strong induction every $n \geq 2$ is a product of primes. □

No separate base case was needed: for $n = 2$ the IH is vacuous and Case 1 applies.

Example 8 — Fibonacci bound, two base cases

Self Tutor

Let $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$. Prove that $f_n \leq 2^{n-1}$ for all $n \geq 1$.

Base cases: $f_1 = 1 \leq 2^0$ and $f_2 = 1 \leq 2^1$.

Step: let $n \geq 3$ and suppose $f_k \leq 2^{k-1}$ for all $1 \leq k < n$. Since $n-1 \geq 1$ and $n-2 \geq 1$ the IH applies to both:

$$f_n = f_{n-1} + f_{n-2} \leq 2^{n-2} + 2^{n-3} \leq 2^{n-2} + 2^{n-2} = 2^{n-1}.$$

By strong induction, $f_n \leq 2^{n-1}$ for all $n \geq 1$. □

Two base cases, because the step uses $n-2$ and is therefore only valid from $n = 3$.

Example 9 — stamps: every $n \geq 12$ is $4a + 5b$

Self Tutor

Prove that every $n \in \mathbb{N}$ with $n \geq 12$ can be written as $n = 4a + 5b$ with $a, b \in \mathbb{N}$.

Base cases: $12 = 4 \cdot 3$, $13 = 4 \cdot 2 + 5$, $14 = 4 + 5 \cdot 2$, $15 = 5 \cdot 3$.

Step: let $n \geq 16$ and suppose the claim for all $12 \leq k < n$. Then $12 \leq n-4 < n$, so $n-4 = 4a + 5b$ for some $a, b \in \mathbb{N}$, and $n = 4(a+1) + 5b$.

By strong induction the claim holds for all $n \geq 12$. □

Example 10 — the division theorem, existence part

Self Tutor

Let $b \geq 1$. Prove that for every $a \in \mathbb{N}$ there exist $q, r \in \mathbb{N}$ with $a = qb + r$ and $0 \leq r < b$.

By strong induction on a . Let $a \in \mathbb{N}$ and suppose the claim for all $a' < a$.

If $a < b$: take $q = 0, r = a$. If $a \geq b$: then $0 \leq a - b < a$, so by the IH $a - b = q'b + r$ with $0 \leq r < b$; hence $a = (q' + 1)b + r$. $\square$

Uniqueness: if $qb + r = q'b + r'$ with $0 \leq r, r' < b$ then $b \mid r - r'$ and $|r - r'| < b$, so $r = r'$, then $q = q'$.

Extend to negative a via $-a$.

Example 11 — base- b expansions exist

Self Tutor

Let $b \geq 2$. Prove that every $n \geq 1$ has a base- b expansion.

Strong induction on n . Let $n \geq 1$ and assume all $1 \leq k < n$ have expansions. Write $n = qb + d$ with $0 \leq d < b$ (division theorem). If $q = 0$ then $n = d$ is a one-digit expansion. If $q \geq 1$ then $1 \leq q < n$ (as $b \geq 2$), so $q = (d_r \dots d_0)_b$ by the IH, and then $n = qb + d = (d_r \dots d_0 d)_b$. $\square$

Well-ordering principle

Every inhabited subset of $\mathbb{N}$ has a least element. This is equivalent to induction, and gives the **minimal counterexample** method: to prove $\forall n \in \mathbb{N}, p(n)$, suppose not; then $\{n \in \mathbb{N} \mid \neg p(n)\}$ is inhabited, so it has a least element m ; derive a contradiction by producing a smaller counterexample (or by checking m directly if m is small).

Example 12 — well-ordering

Self Tutor

Prove that there is no integer n with $0 < n < 1$.

Suppose, for a contradiction, that $S = \{n \in \mathbb{Z} \mid 0 < n < 1\}$ is inhabited. Then $S \subseteq \mathbb{N}$, so by well-ordering S has a least element m . From $0 < m < 1$ we get, multiplying by $m > 0$, $0 < m^2 < m < 1$. So $m^2 \in S$ and $m^2 < m$, contradicting the minimality of m . $\square$

Watch out

- Count the base cases: if the step uses $n - d$, you need base cases $b, b + 1, \dots, b + d - 1$ and the step starts at $n = b + d$. Always check that the smaller value you feed into the IH is still $\geq b$.
- In the step, the IH is “ $p(k)$ for all $b \leq k < n$ ”, and you prove $p(n)$ (not $p(n + 1)$). Mixing the two conventions produces off-by-one errors.
- “ n is not prime, so $n = ab$ ”: say why $1 < a, b < n$. Without that the IH does not apply.

EXERCISES 4.3 and 4.E (strong induction)

Essential: 4.3: 8, 9, 14, 15; 4.E: 13, 19, 26, 27, 29, 30, 31, 32 **Recommended:** 4.3: 10, 18; 4.E: 28, 34, 35

Skills: recognising when weak induction is not enough; proofs with several base cases (Fibonacci-type recurrences, $n - d$ steps); existence of factorisations/expansions; proofs via the well-ordering principle and minimal counterexamples.

C BOOK-LEVEL EXAMPLES

The hardest Chapter 4 exercises need algebra that must be *aimed* at the induction hypothesis, a recurrence that reaches back two steps, or a statement about pictures or sets rather than formulas. The invention is usually the choice of $p(n)$ or the rearrangement that exposes the hypothesis.

Example 13 — sum of squares

Book level

Prove that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Technique: factor out what the target contains, here $(n+1)$, before expanding.

Base case $n = 0$: both sides are 0. Induction step: let $n \in \mathbb{N}$ and assume the formula for n . Then

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)[n(2n+1) + 6(n+1)]}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}, \end{aligned}$$

which is the formula for $n+1$, since $2(n+1)+1 = 2n+3$. By induction the formula holds for all $n \in \mathbb{N}$.

□

Example 14 — Cassini's identity: rearranging to expose the hypothesis

Book level

With $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$, prove that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ for all $n \geq 1$.

Base case $n = 1$: $f_2f_0 - f_1^2 = 0 - 1 = -1 = (-1)^1$.

Induction step: let $n \geq 1$ and assume $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$. Then

$$\begin{aligned} f_{n+2}f_n - f_{n+1}^2 &= (f_{n+1} + f_n)f_n - f_{n+1}^2 \quad \{\text{recurrence}\} \\ &= f_n^2 - f_{n+1}(f_{n+1} - f_n) = f_n^2 - f_{n+1}f_{n-1} \quad \{f_{n+1} - f_n = f_{n-1} \text{ since } n \geq 1\} \\ &= -(f_{n+1}f_{n-1} - f_n^2) = -(-1)^n = (-1)^{n+1}. \quad \{\text{IH}\} \end{aligned}$$

By induction the identity holds for all $n \geq 1$.

□

Example 15 — Binet's formula by strong induction

Book level

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Prove that $f_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ for all $n \in \mathbb{N}$.

Invention: φ and ψ are the roots of $x^2 = x + 1$, so their powers satisfy the Fibonacci recurrence.

First, $\varphi^2 = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} = \varphi + 1$, and similarly $\psi^2 = \psi + 1$. Multiplying by φ^n : $\varphi^{n+2} = \varphi^{n+1} + \varphi^n$; likewise for ψ .

Base cases. $n = 0$: $(1-1)/\sqrt{5} = 0 = f_0$. $n = 1$: $(\varphi - \psi)/\sqrt{5} = \sqrt{5}/\sqrt{5} = 1 = f_1$.

Step. Let $n \geq 2$ and assume the formula for all $k < n$, in particular for $n-1$ and $n-2$ (both ≥ 0). Then

$$f_n = f_{n-1} + f_{n-2} = \frac{(\varphi^{n-1} + \varphi^{n-2}) - (\psi^{n-1} + \psi^{n-2})}{\sqrt{5}} = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

By strong induction the formula holds for all $n \in \mathbb{N}$.

□

Example 16 — a power of two times an odd number

Book level

Prove that every integer $k \geq 1$ can be written as $k = 2^m(2n+1)$ with $m, n \in \mathbb{N}$.

Invention: strong induction, because the smaller case is $k/2$, not $k-1$.

Let $k \geq 1$ and assume the claim for all j with $1 \leq j < k$. If k is odd, then $k = 2n+1$ for some $n \in \mathbb{N}$, and $k = 2^0(2n+1)$. If k is even, then $k = 2j$ with $1 \leq j < k$; by the hypothesis $j = 2^m(2n+1)$, so $k = 2^{m+1}(2n+1)$. By strong induction the claim holds for all $k \geq 1$. $\square$ (Uniqueness of m, n : Example 3.16.)

Example 17 — a statement about pictures

Book level

Say n lines in the plane are in *general position* if no two are parallel and no three pass through one point. Prove that n lines in general position divide the plane into $\frac{n^2+n+2}{2}$ regions.

Invention: find the recurrence $R(n+1) = R(n) + (n+1)$ and prove that in the step.

Let $p(n)$ be: “any n lines in general position cut the plane into $\frac{n^2+n+2}{2}$ regions”. Base case $n=0$: no lines, one region, and $\frac{0+0+2}{2} = 1$.

Step: let $n \in \mathbb{N}$, assume $p(n)$, and take $n+1$ lines in general position. Remove one of them, ℓ . The remaining n lines are in general position, so by $p(n)$ they give $\frac{n^2+n+2}{2}$ regions. Now put ℓ back: it meets each of the other n lines in exactly one point (no two parallel), and these n points are distinct (no three concurrent), so they cut ℓ into $n+1$ pieces. Each piece splits one existing region into two, adding exactly $n+1$ regions. Hence the count is

$$\frac{n^2+n+2}{2} + (n+1) = \frac{n^2+3n+4}{2} = \frac{(n+1)^2 + (n+1) + 2}{2},$$

which is $p(n+1)$. By induction $p(n)$ holds for all $n \in \mathbb{N}$. $\square$

Example 18 — well-ordering implies induction

Book level

Assume the well-ordering principle. Let $p(n)$ be a statement with $p(0)$ true and $p(n) \Rightarrow p(n+1)$ for all $n \in \mathbb{N}$. Prove that $p(n)$ holds for all $n \in \mathbb{N}$.

Invention: look at the least counterexample.

Suppose, for a contradiction, that $S = \{n \in \mathbb{N} \mid \neg p(n)\}$ is inhabited. By well-ordering S has a least element m . Since $p(0)$ holds, $m \neq 0$, so $m-1 \in \mathbb{N}$. As $m-1 < m$ and m is least in S , $m-1 \notin S$, i.e. $p(m-1)$ holds. By the induction step $p(m)$ holds, contradicting $m \in S$. Hence $S = \emptyset$. $\square$

Example 19 — infinite descent

Book level

Use the well-ordering principle to prove that there are no integers $a, b \geq 1$ with $a^2 = 2b^2$.

Invention: among all solutions pick one with the smallest b , then build a smaller one.

Suppose solutions exist, and let $S = \{b \geq 1 \mid \exists a \geq 1, a^2 = 2b^2\}$, an inhabited subset of $\mathbb{N}$. Let b be its least element and fix $a \geq 1$ with $a^2 = 2b^2$. Then a^2 is even, so a is even by the Lemma in Chapter 0; write $a = 2c$ with $c \geq 1$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$. Thus (b, c) is again a solution, so $c \in S$. But $c^2 = b^2/2 < b^2$ gives $c < b$, contradicting the minimality of b . $\square$

This is the “infinite descent” of the book’s title, and a second proof that $\sqrt{2}$ is irrational.

D STRETCH EXERCISES**STRETCH EXERCISES 4 · solutions in Appendix S**

S4.1 Prove that $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$ for all $n \in \mathbb{N}$.

S4.2 Prove that $6 \mid n^3 - n$ for all $n \in \mathbb{N}$. [Hint: $(n+1)^3 - (n+1) = (n^3 - n) + 3n(n+1)$]

S4.3 Define $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for $0 \leq k \leq n$ and $\binom{n}{k} = 0$ for $k > n$. Prove Pascal's rule $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$ from this formula, and then prove by induction on n that $\binom{n}{k} \in \mathbb{N}$ for all k .

S4.4 Prove that for every $n \geq 1$, a $2^n \times 2^n$ board with one square removed can be tiled by L-shaped pieces of three squares. [Hint: split into four quadrants and place one piece at the centre]

S4.5 Prove that for all $n \in \mathbb{N}$ there is no integer $d \geq 2$ dividing both f_n and f_{n+1} . [Hint: if $d \mid f_{n+1}$ and $d \mid f_{n+2}$ then $d \mid f_n$]

S4.6 Prove that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$ for all $n \geq 1$. [Hint: the gap inequality is $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$]

S4.7 Prove that every $n \geq 1$ is a sum of distinct powers of 2. [Hint: strong induction; subtract the largest power of 2 not exceeding n]

S4.8 Prove that $\sum_{k=1}^n f_k = f_{n+2} - 1$ for all $n \geq 1$.

S4.9 Prove that every $n \geq 8$ can be written as $3a + 5b$ with $a, b \in \mathbb{N}$. [Hint: how many base cases does a step of size 3 need?]

E REVIEW SET 4

- Weak: $p(b)$ and $\forall n \geq b, p(n) \Rightarrow p(n+1)$.
- Strong: $\forall n \geq b, (\forall k, b \leq k < n \Rightarrow p(k)) \Rightarrow p(n)$; base cases as many as the step needs.
- Template words: “Let $p(n)$ be”, “Base case”, “Induction step”, “(IH)”, “By induction”.
- Sums: $\sum_1^{n+1} = \sum_1^n + a_{n+1}$. Divisibility: $x_{n+1} = c \cdot x_n +$ (multiple). Inequalities: IH then a gap inequality.
- Binomial theorem $(x+y)^n = \sum \binom{n}{k} x^k y^{n-k}$; Pascal's rule.
- Well-ordering: inhabited $S \subseteq \mathbb{N}$ has a least element; minimal counterexample.

Chapter 5

Relations

Contents

- A** Relations and their properties (§5.1)
- B** Equivalence relations and partitions (§5.2)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 5

Planning

Essential: 5.1: 6, 13, 22, 23, 28, 39, 40, 41; 5.2: 24, 27, 33; 5.E: 18, 22, 23
Recommended: 5.1: 10, 11, 18, 24, 29, 33, 34; 5.2: 25, 28, 30, 34, 36
 Tested in **Quiz 3** (21 Oct) together with modular arithmetic (§7.3).

A RELATIONS AND THEIR PROPERTIES

Relation

A **relation** R from a set X to a set Y is a statement $a R b$ about pairs (a, b) with $a \in X, b \in Y$; equivalently, it is its **graph** $\text{Gr}(R) = \{(a, b) \in X \times Y \mid a R b\} \subseteq X \times Y$. A relation *on* X is a relation from X to X . Every subset of $X \times Y$ is the graph of exactly one relation, so a set with n elements has 2^{n^2} relations on it. Examples on $\mathbb{Z}$: $=, \leq, <, |, "a \equiv b \pmod{n}"$ (i.e. $n \mid a - b$), “ a and b have the same parity”. On $\mathcal{P}(X)$: $\subseteq$. Between X and Y : “ $f(a) = b$ ” for a function $f : X \rightarrow Y$ (its graph is $\text{Gr}(f)$).

Properties of a relation R on X

R is ...	definition	to disprove, exhibit ...
reflexive	$\forall a \in X, a R a$	one a with $\neg(a R a)$
symmetric	$\forall a, b \in X, (a R b \Rightarrow b R a)$	a, b with $a R b$ and $\neg(b R a)$
transitive	$\forall a, b, c \in X, (a R b \wedge b R c \Rightarrow a R c)$	a, b, c with $a R b, b R c, \neg(a R c)$
antisymmetric	$\forall a, b \in X, (a R b \wedge b R a \Rightarrow a = b)$	$a \neq b$ with $a R b$ and $b R a$

relation	reflexive	symmetric	transitive	antisymmetric
$=$ on any set	✓	✓	✓	✓
$\leq$ on $\mathbb{R}$; $\subseteq$ on $\mathcal{P}(X)$; $ $ on $\mathbb{N}$	✓	✗	✓	✓
$<$ on $\mathbb{R}$	✗	✗	✓	✓ (vacuously)
$\equiv \pmod{n}$ on $\mathbb{Z}$	✓	✓	✓	✗ ($n \geq 2$)
$\neq$ on a set with ≥ 2 elements	✗	✓	✗	✗

Proving properties

Each property is a $\forall$ -statement, usually with an implication inside; write exactly that.

- Reflexive: “Let $a \in X$. Then ..., so $a R a$.”
- Symmetric: “Let $a, b \in X$ and suppose $a R b$ Hence $b R a$.”
- Transitive: “Let $a, b, c \in X$ and suppose $a R b$ and $b R c$ Hence $a R c$.”

- Antisymmetric: “Let $a, b \in X$ and suppose $a R b$ and $b R a$ Hence $a = b$.”
Unfold “ $a R b$ ” into its definition on the first line (“so $3 \mid a - b$; fix k with $a - b = 3k$ ”).

Example 1 — congruence modulo 3

Self Tutor

On $\mathbb{Z}$ define $a \sim b$ to mean $3 \mid a - b$. Prove that $\sim$ is reflexive, symmetric and transitive, and show that it is not antisymmetric.

Reflexive. Let $a \in \mathbb{Z}$. Then $a - a = 0 = 3 \cdot 0$, so $3 \mid a - a$, i.e. $a \sim a$.

Symmetric. Let $a, b \in \mathbb{Z}$ with $a \sim b$; fix $k \in \mathbb{Z}$ with $a - b = 3k$. Then $b - a = 3(-k)$, so $3 \mid b - a$, i.e. $b \sim a$.

Transitive. Let $a, b, c \in \mathbb{Z}$ with $a \sim b$ and $b \sim c$; fix $k, l \in \mathbb{Z}$ with $a - b = 3k$ and $b - c = 3l$. Then $a - c = (a - b) + (b - c) = 3(k + l)$, so $a \sim c$.

Not antisymmetric. $0 \sim 3$ and $3 \sim 0$ (both differences are ± 3), but $0 \neq 3$. □

Example 2 — divisibility is antisymmetric on $\mathbb{N}$ but not on $\mathbb{Z}$

Self Tutor

Prove that $\mid$ is antisymmetric on $\mathbb{N}$. Show that it is not antisymmetric on $\mathbb{Z}$.

Let $a, b \in \mathbb{N}$ with $a \mid b$ and $b \mid a$; fix $k, l \in \mathbb{N}$ with $b = ka$ and $a = lb$ (the quotients are ≥ 0 since $a, b \geq 0$). Then $a = lka$. If $a = 0$ then $b = k \cdot 0 = 0 = a$. If $a \neq 0$ then $lk = 1$, and since $k, l \in \mathbb{N}$ this forces $k = l = 1$, so $a = b$.

On $\mathbb{Z}$: $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$. □

Example 3 — a relation with some properties and not others

Self Tutor

On $\mathbb{Z}$ define $a R b$ to mean that $a + b$ is odd. Decide whether R is reflexive, symmetric, transitive.

Not reflexive: $0 + 0 = 0$ is even, so $\neg(0 R 0)$.

Symmetric: let $a, b \in \mathbb{Z}$ with $a R b$. Then $b + a = a + b$ is odd, so $b R a$.

Not transitive: $1 R 2$ and $2 R 3$ (sums 3 and 5), but $1 + 3 = 4$ is even, so $\neg(1 R 3)$. □

Relations that are functions

A relation R from X to Y is (the graph of) a function $X \rightarrow Y$ iff $\forall a \in X, \exists! b \in Y, a R b$: every input is related to exactly one output. Then $f(a)$ is that b .

Example 4 — is it a function?

Self Tutor

Which of the relations $G_1 = \{(x, y) \in \mathbb{R}^2 \mid y = x^2\}$ and $G_2 = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$ is the graph of a function $\mathbb{R} \rightarrow \mathbb{R}$?

G_1 : for each $x \in \mathbb{R}$ there is exactly one y with $y = x^2$. So G_1 is the graph of $x \mapsto x^2$.

G_2 : for $x = 1$ both $(1, 1)$ and $(1, -1)$ lie in G_2 (not unique), and for $x = -1$ there is no y with $y^2 = -1$ (not existent). So G_2 is not the graph of a function. □

Watch out

- Reflexivity is about *all* elements. “Symmetric and transitive imply reflexive” is false: the empty relation on $\{1\}$ is symmetric and transitive but not reflexive (the argument “ $a R b$, so $b R a$, so $a R a$ ” needs some b to exist).

- Antisymmetric is not the negation of symmetric: $=$ is both; $\neq$ is neither.
- To refute a property one concrete counterexample is enough, but you must check it: state the pairs and why they are (not) related.

EXERCISES 5.1

Essential: 6, 13, 22, 23, 28, 39, 40, 41 **Recommended:** 10, 11, 18, 24, 29, 33, 34

Skills: writing a relation as a set of pairs and back; deciding and proving/refuting reflexive, symmetric, transitive, antisymmetric for concrete relations on $\mathbb{Z}$, $\mathbb{R}$, $\mathcal{P}(X)$, finite sets; relations as functions (the $\forall\exists!$ test); counting relations.

B EQUIVALENCE RELATIONS AND PARTITIONS**Equivalence relation, class, quotient**

An **equivalence relation** on X is a relation $\sim$ that is reflexive, symmetric and transitive. For $a \in X$, the **equivalence class** of a is

$$[a]_{\sim} = \{b \in X \mid a \sim b\} \subseteq X,$$

and the **quotient** of X by $\sim$ is the set of all classes, $X/\sim = \{[a]_{\sim} \mid a \in X\} \subseteq \mathcal{P}(X)$. Membership rule: $b \in [a] \Leftrightarrow a \sim b$.

Lemma: classes are equal or disjoint

Let $\sim$ be an equivalence relation on X and $a, b \in X$. Then

$$a \sim b \Leftrightarrow [a] = [b] \Leftrightarrow [a] \cap [b] \text{ is inhabited.}$$

Moreover $a \in [a]$ for every a , so every class is inhabited and every element lies in some class.

Example 5 — proof of the Lemma**Self Tutor**

Prove the Lemma.

$(a \sim b \Rightarrow [a] = [b])$ Suppose $a \sim b$. Let $c \in [a]$, i.e. $a \sim c$. By symmetry $b \sim a$, and by transitivity $(b \sim a, a \sim c) \Rightarrow b \sim c$, i.e. $c \in [b]$. So $[a] \subseteq [b]$; by symmetry of the situation $(b \sim a)$ also $[b] \subseteq [a]$.

$([a] = [b] \Rightarrow [a] \cap [b] \text{ inhabited})$ $a \in [a]$ by reflexivity, and $[a] = [b]$, so $a \in [a] \cap [b]$.

$([a] \cap [b] \text{ inhabited} \Rightarrow a \sim b)$ Fix $c \in [a] \cap [b]$: $a \sim c$ and $b \sim c$. By symmetry $c \sim b$; by transitivity $a \sim b$. $\square$

Partition

A **partition** of X is a set $\mathcal{U}$ of subsets of X such that (i) every $U \in \mathcal{U}$ is inhabited; (ii) distinct members are disjoint: $U, V \in \mathcal{U}, U \neq V \Rightarrow U \cap V = \emptyset$; (iii) they cover X : $\bigcup_{U \in \mathcal{U}} U = X$ (every $a \in X$ lies in some $U \in \mathcal{U}$).

Equivalently: every $a \in X$ lies in *exactly one* member of $\mathcal{U}$.

Fundamental theorem: equivalence relations = partitions

1 If $\sim$ is an equivalence relation on X , then $X/\sim$ is a partition of X (by the Lemma).

2 If $\mathcal{U}$ is a partition of X , then $a \sim_{\mathcal{U}} b \Leftrightarrow \exists U \in \mathcal{U}, (a \in U \wedge b \in U)$ defines an equivalence relation

on X whose classes are exactly the members of $\mathcal{U}$.

3 These constructions are inverse to each other: equivalence relations on X correspond bijectively to partitions of X .

Consequence for computations: to describe $X/\sim$, find one representative per class and prove (a) every element is equivalent to one of the representatives, (b) distinct representatives are not equivalent.

Example 6 — $\mathbb{Z}/n\mathbb{Z}$ has exactly n elements

Self Tutor

Let $n \geq 1$ and let $a \equiv b \pmod{n}$ mean $n \mid a - b$ (an equivalence relation, as in Example 5.1). Write $[a]_n$ for the class of a and $\mathbb{Z}/n\mathbb{Z}$ for the quotient. Prove that $\mathbb{Z}/n\mathbb{Z} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$ and that these n classes are distinct.

(a) Let $a \in \mathbb{Z}$. By the division theorem $a = qn + r$ with $0 \leq r < n$. Then $a - r = qn$, so $a \equiv r$, so $[a]_n = [r]_n$ by the Lemma, with $r \in \{0, \dots, n-1\}$.

(b) Let $0 \leq r < s \leq n-1$. Then $0 < s - r < n$, so $n \nmid s - r$ (the only multiple of n in $(0, n)$ would be $\geq n$). Hence $r \not\equiv s$ and $[r]_n \neq [s]_n$ by the Lemma.

So $\mathbb{Z}/n\mathbb{Z}$ consists of exactly the n distinct classes $[0]_n, \dots, [n-1]_n$; the class of a is the class of its remainder. $\square$

Example 7 — describing the classes

Self Tutor

On $\mathbb{R}$ define $x \sim y$ iff $x - y \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation, describe $[x]$, and show that every class contains exactly one element of $[0, 1)$.

Reflexive: $x - x = 0 \in \mathbb{Z}$. *Symmetric:* if $x - y \in \mathbb{Z}$ then $y - x = -(x - y) \in \mathbb{Z}$. *Transitive:* if $x - y, y - z \in \mathbb{Z}$ then $x - z = (x - y) + (y - z) \in \mathbb{Z}$.

$[x] = \{y \in \mathbb{R} \mid x - y \in \mathbb{Z}\} = \{x + k \mid k \in \mathbb{Z}\}$.

Exactly one representative in $[0, 1)$: existence: $r = x - \lfloor x \rfloor \in [0, 1)$ and $x - r = \lfloor x \rfloor \in \mathbb{Z}$, so $r \in [x]$.

Uniqueness: if $r, r' \in [x] \cap [0, 1)$ then $r - r' \in \mathbb{Z}$ (transitivity/symmetry) and $|r - r'| < 1$, so $r = r'$. $\square$

Example 8 — fractions: transitivity needs care

Self Tutor

On $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ define $(a, b) \sim (c, d)$ iff $ad = bc$. Prove that $\sim$ is transitive. (Its classes are the rational numbers: $[(a, b)]$ “is” $\frac{a}{b}$.)

Let $(a, b), (c, d), (e, f) \in X$ with $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, i.e. $ad = bc$ and $cf = de$. We need $af = be$. Multiply the first equation by f and use the second:

$$adf = bcf = b(cf) = bde, \quad \text{so} \quad d(af - be) = 0.$$

Since $d \neq 0$ (second coordinates are non-zero), $af - be = 0$, i.e. $af = be$. Hence $(a, b) \sim (e, f)$. $\square$

Example 9 — from partition to relation and back

Self Tutor

Let $X = \{1, 2, 3, 4, 5\}$ and $\mathcal{U} = \{\{1, 3\}, \{2\}, \{4, 5\}\}$. Write the corresponding equivalence relation as a set of pairs. How many equivalence relations are there on $\{1, 2, 3\}$?

$a \sim b$ iff a, b lie in the same member: $\text{Gr}(\sim) = \{(1, 1), (3, 3), (1, 3), (3, 1), (2, 2), (4, 4), (5, 5), (4, 5), (5, 4)\}$ (nine pairs: $2^2 + 1^2 + 2^2$).

Equivalence relations on $\{1, 2, 3\}$ correspond to partitions: $\{\{1, 2, 3\}\}$; $\{\{1\}, \{2\}, \{3\}\}$; and three of the form $\{\{a, b\}, \{c\}\}$. Total: 5. $\square$

The quotient function

For an equivalence relation $\sim$ on X , the **quotient function** is $q_\sim : X \rightarrow X/\sim$, $q_\sim(a) = [a]_\sim$. It is surjective (every class is $[a]$ for some a), and $q_\sim(a) = q_\sim(b) \Leftrightarrow a \sim b$ (Lemma).

For $\equiv \pmod{n}$: $q_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ sends a to $[r]_n$ where r is the remainder of a on division by n (Example 5.6); so $q_n(a) = q_n(b)$ iff a and b have the same remainder.

Theorem 5.2.35: every surjection is a quotient

Let $p : X \rightarrow Z$ be a surjection. Define $a \sim b \Leftrightarrow p(a) = p(b)$. Then $\sim$ is an equivalence relation on X , and $f : X/\sim \rightarrow Z$, $f([a]) = p(a)$, is a well-defined bijection with $f \circ q_\sim = p$. The relation $\sim$ is the only equivalence relation on X for which such a bijection exists.

Example 10 — the induced bijection (proof of the core of Theorem 5.2.35)**Self Tutor**

Let $p : X \rightarrow Z$ be surjective and $a \sim b \Leftrightarrow p(a) = p(b)$. Prove that $\sim$ is an equivalence relation and that $f([a]) = p(a)$ defines a bijection $X/\sim \rightarrow Z$.

Equivalence relation. Reflexive: $p(a) = p(a)$. Symmetric: $p(a) = p(b) \Rightarrow p(b) = p(a)$. Transitive: $p(a) = p(b)$ and $p(b) = p(c)$ give $p(a) = p(c)$.

Well-defined. If $[a] = [b]$ then $a \sim b$ (Lemma), i.e. $p(a) = p(b)$: the proposed value does not depend on the representative.

Injective. Suppose $f([a]) = f([b])$, i.e. $p(a) = p(b)$. Then $a \sim b$, so $[a] = [b]$ by the Lemma.

Surjective. Let $z \in Z$. Since p is surjective, fix $a \in X$ with $p(a) = z$; then $f([a]) = z$. □

Defining a function on a quotient: well-definedness

To define $F : X/\sim \rightarrow Y$ by a formula $F([a]) = h(a)$ you must prove: $\forall a, b \in X, (a \sim b \Rightarrow h(a) = h(b))$. Otherwise the “function” assigns two values to one class. This check is mandatory in every exercise that defines something on $\mathbb{Z}/n\mathbb{Z}$, $X/\sim$, fractions, etc.

Example 11 — well-defined and not well-defined on $\mathbb{Z}/n\mathbb{Z}$ **Self Tutor**

(a) Prove that $[a]_n + [b]_n := [a + b]_n$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$. (b) Show that $F : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}$, $F([a]_n) = a$ is not well-defined (for $n \geq 1$).

(a) Suppose $[a]_n = [a']_n$ and $[b]_n = [b']_n$, i.e. $n \mid a - a'$ and $n \mid b - b'$; fix k, l with $a - a' = nk$, $b - b' = nl$. Then $(a + b) - (a' + b') = n(k + l)$, so $n \mid (a + b) - (a' + b')$, i.e. $[a + b]_n = [a' + b']_n$. The result does not depend on the representatives. □

(b) $[0]_n = [n]_n$ (since $n \mid n - 0$), but the rule gives $F([0]_n) = 0$ and $F([n]_n) = n \neq 0$. So F is not well-defined. □

Watch out

- $[a] = [b]$ does not mean $a = b$; it means $a \sim b$. An equivalence class is a *set*; $X/\sim$ is a set of sets.
- When you define anything on a quotient, write the well-definedness check before anything else. “Choose a representative” must be followed by “and the result is independent of the choice because ...”.
- To prove a relation is an equivalence relation you need all three properties, each with its own “Let ...” line. Reflexivity is the one people forget.
- Counting classes: existence of a representative for every element *and* distinctness of the representatives, both proved.

EXERCISES 5.2 and 5.E**Essential:** 5.2: 24, 27, 33; 5.E: 18, 22, 23 **Recommended:** 5.2: 25, 28, 30, 34, 36

Skills: proving a relation is an equivalence relation; computing and describing equivalence classes and quotients ($\mathbb{Z}/n\mathbb{Z}$, $\mathbb{R}/\mathbb{Z}$ -type examples, fractions, functions grouped by a value); converting between partitions and equivalence relations; the quotient function and Theorem 5.2.35 (surjections $\leftrightarrow$ quotients, Exercise 5.2.33 on remainders); proving well-definedness of functions and operations on quotients; using Lemma 5.2.29-type bijections class by class (Exercise 5.2.30).

C BOOK-LEVEL EXAMPLES

The hardest Chapter 5 exercises ask you to count, to combine relations, to recognise a quotient as the kernel of a function, or to build a function out of a quotient and prove it is well-defined. The exercise from the book page you have (5.2.30) is worked in full below.

Example 12 — counting relations**Book level**

Let X be a set with n elements. How many relations on X are there? How many of them are reflexive? How many are symmetric? Justify your answers.

Invention: a relation is a subset of $X \times X$, so count subsets with constraints.

$X \times X$ has n^2 elements, so it has 2^{n^2} subsets; every subset is the graph of exactly one relation, so there are 2^{n^2} relations.

Reflexive: the n pairs (a, a) must be in the graph; the other $n^2 - n$ pairs are free. So 2^{n^2-n} .

Symmetric: the graph is determined by which of the n pairs (a, a) it contains and which of the $\binom{n}{2} = \frac{n(n-1)}{2}$ unordered pairs $\{a, b\}$ with $a \neq b$ it contains (it contains (a, b) iff it contains (b, a)). So $2^{n + n(n-1)/2} = 2^{n(n+1)/2}$. $\square$

Example 13 — combining equivalence relations**Book level**

Let $\sim$ and $\approx$ be equivalence relations on X . Define $a R b$ to mean $a \sim b$ and $a \approx b$. Prove that R is an equivalence relation. Show by example that “ $a \sim b$ or $a \approx b$ ” need not be one.

Reflexive: $a \sim a$ and $a \approx a$, so $a R a$. *Symmetric:* if $a \sim b$ and $a \approx b$, then $b \sim a$ and $b \approx a$. *Transitive:* if $a R b$ and $b R c$, then $a \sim b$, $b \sim c$ give $a \sim c$, and $a \approx b$, $b \approx c$ give $a \approx c$; so $a R c$. $\square$

The “or” relation can fail transitivity. On $X = \{1, 2, 3\}$ let $\sim$ have classes $\{1, 2\}, \{3\}$ and $\approx$ have classes $\{1\}, \{2, 3\}$. Then $1 \sim 2$ and $2 \approx 3$, so 1 and 2 are related, and so are 2 and 3; but $1 \not\sim 3$ and $1 \not\approx 3$, so 1 and 3 are not.

Example 14 — a quotient of the plane**Book level**

On $\mathbb{R}^2$ define $(x, y) \sim (x', y')$ iff $x^2 + y^2 = x'^2 + y'^2$. Prove that $\sim$ is an equivalence relation, describe its classes, and show that $\mathbb{R}^2/\sim$ is in bijection with $[0, \infty)$.

Invention: recognise $\sim$ as “same value under $p(x, y) = x^2 + y^2$ ” and use Theorem 5.2.35.

Let $p: \mathbb{R}^2 \rightarrow [0, \infty)$, $p(x, y) = x^2 + y^2$. Then $(x, y) \sim (x', y')$ iff $p(x, y) = p(x', y')$, which is an equivalence relation by the first part of Example 5.10.

The class of (x, y) is $\{(x', y') \mid x'^2 + y'^2 = r^2\}$ with $r = \sqrt{x^2 + y^2}$: the circle of radius r about the origin, or $\{(0, 0)\}$ when $r = 0$.

p is surjective: for $t \geq 0$, $p(\sqrt{t}, 0) = t$. By Theorem 5.2.35, $[(x, y)] \mapsto x^2 + y^2$ is a well-defined bijection $\mathbb{R}^2/\sim \rightarrow [0, \infty)$. $\square$

Example 15 — pulling back an equivalence relation

Book level

Let $f : X \rightarrow Y$ be a function and $\sim$ an equivalence relation on Y . Define $a \approx b$ iff $f(a) \sim f(b)$. Prove that $\approx$ is an equivalence relation on X .

Reflexive: $f(a) \sim f(a)$ by reflexivity of $\sim$. *Symmetric:* if $f(a) \sim f(b)$ then $f(b) \sim f(a)$. *Transitive:* if $f(a) \sim f(b)$ and $f(b) \sim f(c)$ then $f(a) \sim f(c)$. $\square$

With $\sim$ equal to $=$ on Y this is the kernel relation of Theorem 5.2.35; the previous example is the case $f = p$.

Example 16 — Exercise 5.2.30: gluing bijections class by class

Book level

Let $\sim$ and $\approx$ be equivalence relations on X and Y . Suppose $p : X/\sim \rightarrow Y/\approx$ is a bijection, and that for each class $E \in X/\sim$ a bijection $h_E : E \rightarrow p(E)$ is given. Prove that there is a bijection $h : X \rightarrow Y$.

Invention: define h piecewise, $h(a) = h_{[a]}(a)$, and use that distinct classes are disjoint.

Define $h : X \rightarrow Y$ by $h(a) = h_{[a]}(a)$. This makes sense: $a \in [a]$, $h_{[a]}$ is a function $[a] \rightarrow p([a])$, and $p([a]) \subseteq Y$.

Injective. Let $a, b \in X$ with $h(a) = h(b)$, i.e. $h_{[a]}(a) = h_{[b]}(b)$; call this element y . Then $y \in p([a])$ and $y \in p([b])$, so these two classes of $\approx$ intersect, hence are equal (Lemma). Since p is injective, $[a] = [b]$. Now $h_{[a]}(a) = h_{[a]}(b)$ with $h_{[a]}$ injective, so $a = b$.

Surjective. Let $y \in Y$ and put $F = [y]_{\approx} \in Y/\approx$. Since p is surjective, fix $E \in X/\sim$ with $p(E) = F$. Since $h_E : E \rightarrow F$ is surjective and $y \in F$, fix $a \in E$ with $h_E(a) = y$. As $a \in E$ and E is a class, $E = [a]$; so $h(a) = h_{[a]}(a) = h_E(a) = y$. $\square$

Example 17 — well-defined or not, with a twist

Book level

Let $n \geq 1$. (a) Prove that $[a]_n \cdot [b]_n := [ab]_n$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$. (b) Deduce that $S([a]_n) = [a^2]_n$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. (c) Show that $T([a]_3) = [2^a]_3$ is *not* well-defined on $\mathbb{Z}/3\mathbb{Z}$.

Invention: write the difference of the two candidate outputs as a multiple of n .

(a) Suppose $[a] = [a']$ and $[b] = [b']$, so $a - a' = nk$ and $b - b' = nl$ for some $k, l \in \mathbb{Z}$. Then $ab - a'b' = a(b - b') + b'(a - a') = n(al + b'k)$, so $[ab] = [a'b']$.

(b) Take $b = a$ and $b' = a'$ in (a): if $[a] = [a']$ then $[a^2] = [a'^2]$.

(c) $[0]_3 = [3]_3$, but $2^0 = 1$ and $2^3 = 8$, and $8 - 1 = 7$ is not a multiple of 3, so $[2^0]_3 \neq [2^3]_3$. The rule assigns two different values to one class. $\square$

Example 18 — functions out of a quotient

Book level

Let $\sim$ be an equivalence relation on X with quotient function $q : X \rightarrow X/\sim$, and let Y be a set. Let C be the set of functions $h : X \rightarrow Y$ such that $a \sim b \Rightarrow h(a) = h(b)$ for all $a, b \in X$. Prove that

$$\Phi : \{\text{functions } X/\sim \rightarrow Y\} \rightarrow C, \quad \Phi(g) = g \circ q,$$

is a bijection.

Invention: injectivity is cancellation by the surjection q ; surjectivity is the well-definedness check.

First, $\Phi(g) \in C$: if $a \sim b$ then $[a] = [b]$, so $g(q(a)) = g([a]) = g([b]) = g(q(b))$.

Injective. Suppose $g \circ q = g' \circ q$. Let $E \in X/\sim$; since q is surjective fix a with $q(a) = E$. Then $g(E) = g(q(a)) = g'(q(a)) = g'(E)$. So $g = g'$.

Surjective. Let $h \in C$. Define $g : X/\sim \rightarrow Y$ by $g([a]) = h(a)$; this is well-defined because $[a] = [b]$ implies $a \sim b$, which implies $h(a) = h(b)$. Then $(g \circ q)(a) = g([a]) = h(a)$ for all a , so $\Phi(g) = h$. $\square$

D STRETCH EXERCISES

STRETCH EXERCISES 5 · solutions in Appendix S

- S5.1** On $\mathbb{Z}$ define $a R b$ iff $a \mid b$. Decide, with proof or counterexample, which of reflexive, symmetric, transitive, antisymmetric hold.
- S5.2** Let R be a symmetric and transitive relation on X such that for every $a \in X$ there is some $b \in X$ with $a R b$. Prove that R is reflexive.
- S5.3** On $\mathbb{R}$ define $x \sim y$ iff $x - y \in \mathbb{Q}$. Prove that $\sim$ is an equivalence relation, show that $[0] = \mathbb{Q}$ and $[\sqrt{2}] = \{\sqrt{2} + r \mid r \in \mathbb{Q}\}$, and prove that $[\sqrt{2}] \neq [\sqrt{3}]$. [Hint: $(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2}) = 1$]
- S5.4** How many equivalence relations are there on a set with 4 elements? [Hint: count partitions by shape]
- S5.5** On $\mathcal{P}(\mathbb{N})$ define $U \sim V$ iff $U \Delta V$ is finite. Prove that $\sim$ is an equivalence relation. [Hint: $U \Delta W \subseteq (U \Delta V) \cup (V \Delta W)$]
- S5.6** Let $\sim$ be an equivalence relation on X and $f : X \rightarrow Y$ a function with $a \sim b \Rightarrow f(a) = f(b)$. Let $g : X/\sim \rightarrow Y$ be the induced function $g([a]) = f(a)$. Prove that $g[X/\sim] = f[X]$ and that g is injective if and only if $f(a) = f(b) \Rightarrow a \sim b$ for all a, b .
- S5.7** Let $m, n \geq 1$. Prove that $[a]_n \mapsto [a]_m$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ if and only if $m \mid n$.
- S5.8** Prove that a relation R on X is an equivalence relation if and only if R is reflexive and satisfies: for all $a, b, c \in X$, if $a R b$ and $a R c$ then $b R c$.

E REVIEW SET 5

- Relation on X = subset of $X \times X$; 2^{n^2} of them.
- Reflexive $\forall a a R a$; symmetric $a R b \Rightarrow b R a$; transitive $a R b \wedge b R c \Rightarrow a R c$; antisymmetric $a R b \wedge b R a \Rightarrow a = b$.
- Function $\Leftrightarrow \forall a \exists! b, a R b$.
- Equivalence relation = R+S+T; $[a] = \{b \mid a \sim b\}$; $X/\sim = \{[a]\}$.
- $a \sim b \Leftrightarrow [a] = [b] \Leftrightarrow [a] \cap [b] \neq \emptyset$.
- Partition: inhabited, pairwise disjoint, cover. Equivalence relations $\leftrightarrow$ partitions.
- $\mathbb{Z}/n\mathbb{Z} = \{[0], \dots, [n-1]\}$, class = remainder.
- $q_\sim : X \rightarrow X/\sim$ surjective; every surjection p is $q_\sim$ up to the bijection $[a] \mapsto p(a)$, with $a \sim b \Leftrightarrow p(a) = p(b)$.
- Defining $F([a]) = h(a)$ needs $a \sim b \Rightarrow h(a) = h(b)$.

Appendix T

One-page proof toolbox

Contents

- A** Reading off the first line of a proof
Standard unfoldings

Planning

Use this page while doing the exercises: find the outermost symbol of what you must prove, copy the first line, unfold the definitions, repeat.

A READING OFF THE FIRST LINE OF A PROOF

goal looks like	first line(s)	last line
$\forall x \in X, p(x)$	Let $x \in X$.	... so $p(x)$. Since x was arbitrary, ...
$\exists x \in X, p(x)$	Put $x = \dots$ (then check $x \in X$).	... so $p(x)$.
$\exists! x$	Existence: as above. Uniqueness: suppose x, y both satisfy p .	... so $x = y$.
$P \Rightarrow Q$	Suppose P . (or: We prove the contrapositive; suppose $\neg Q$.)	... hence Q . (... hence $\neg P$.)
$P \wedge Q$	First we prove P Next we prove Q .	both done.
$P \vee Q$	Suppose $\neg P$.	... hence Q .
$P \Leftrightarrow Q$	$(\Rightarrow)$ Suppose P ... $(\Leftarrow)$ Suppose Q ...	both directions done.
$\neg P$, “there is no”, “irrational”, $X = \emptyset$	Suppose, for a contradiction, that P (that such an object exists; that $a \in X$).	... contradiction. Hence $\neg P$.
$X \subseteq Y$	Let $a \in X$.	... so $a \in Y$.
$X = Y$	$(\subseteq)$ Let $a \in X$... $(\supseteq)$ Let $a \in Y$... or a $\Leftrightarrow$ -chain.	both inclusions done.
f injective	Let $a, b \in X$ with $f(a) = f(b)$.	... so $a = b$.
f surjective	Let $b \in Y$. Put $a = \dots$	... so $f(a) = b$.
$\forall n \geq b, p(n)$	Base case $n = b$: ... Induction step: let $n \geq b$, suppose $p(n)$.	... so $p(n+1)$. By induction ...
R reflexive / symmetric / transitive	Let a . / Let a, b with aRb . / Let a, b, c with aRb, bRc .	aRa / bRa / aRc .
$F([a]) = h(a)$ well-defined	Let $a \sim b$.	... so $h(a) = h(b)$.

Using an assumption: $\exists$ or “ $b \mid a$ ” or “ $b \in f[U]$ ” $\rightarrow$ “Fix k with ...”. $P \vee Q$ or “ $a \in X \cup Y$ ” $\rightarrow$ cases. $P \wedge Q$ or “ $a \in X \cap Y$ ” $\rightarrow$ two facts. $\forall \rightarrow$ plug in what you need. $P \Rightarrow Q$ and $P \rightarrow Q$.

Standard unfoldings

$b \mid a$	$\exists q \in \mathbb{Z}, a = qb$	$a \in \bigcap_{i \in I} X_i$	$\forall i \in I, a \in X_i$
n even / odd	$n = 2k$ / $n = 2k + 1, k \in \mathbb{Z}$	$a \in \bigcup_{i \in I} X_i$	$\exists i \in I, a \in X_i$
$x \in \mathbb{Q}$	$x = \frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0$	$(a, b) \in X \times Y$	$a \in X \wedge b \in Y$
$a \in \{x \in X \mid p(x)\}$	$a \in X \wedge p(a)$	$b \in f[U]$	$\exists a \in U, f(a) = b$
$b \in \{e(x) \mid x \in X\}$	$\exists x \in X, b = e(x)$	$a \in f^{-1}[V]$	$f(a) \in V$
$X \subseteq Y$	$\forall a \in X, a \in Y$	$g \circ f = \text{id}_X$	$\forall a \in X, g(f(a)) = a$
$U \subseteq \mathcal{P}(X)$	$U \subseteq X$	$b \in [a]_{\sim}$	$a \sim b$
$a \in X \cap Y$	$a \in X \wedge a \in Y$	$[a] = [b]$	$a \sim b$ (Lemma)
$a \in X \cup Y$	$a \in X \vee a \in Y$	$a \equiv b \pmod{n}$	$n \mid a - b$
$a \in X \setminus Y$	$a \in X \wedge a \notin Y$		

Appendix S

Solutions to the stretch exercises

Contents

- A** Chapter 0
- B** Chapter 1
- C** Chapter 2
- D** Chapter 3
- E** Chapter 4
- F** Chapter 5

Planning

Compare your own attempt line by line: did you introduce every variable, unfold every definition, and say which strategy you were using?

A CHAPTER 0

S0.1 Since $1 < 2 < 10$, $0 < \log_{10} 2 < 1$. Suppose $\log_{10} 2 = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \geq 1$; then $a \geq 1$. By definition of log, $10^{a/b} = 2$, so $2^b = 10^a = 2^a 5^a$. If $b \leq a$, cancelling 2^b gives $1 = 2^{a-b} 5^a \geq 5$, impossible. If $b > a$, cancelling 2^a gives $2^{b-a} = 5^a$: the left side is even ($b - a \geq 1$) and the right side is odd (a product of odd numbers). Contradiction either way.

S0.2 By the division theorem $n = 3k$, $3k + 1$ or $3k + 2$. Then $n^2 + 1$ equals $3(3k^2) + 1$, $3(3k^2 + 2k) + 2$ or $3(3k^2 + 4k + 1) + 2$ respectively, so its remainder on division by 3 is 1 or 2, never 0.

S0.3 Write $n = \sum_i d_i 3^i$ and $s = \sum_i d_i$. Each 3^i is odd (a product of odd numbers), say $3^i = 2m_i + 1$. Then $n - s = \sum_i d_i(3^i - 1) = 2 \sum_i d_i m_i$ is even. So $n = s + 2M$ with $M \in \mathbb{Z}$: if $s = 2j$ then $n = 2(j + M)$; if $n = 2j$ then $s = 2(j - M)$. Hence n is even iff s is even.

S0.4 Fix $k, l \in \mathbb{Z}$ with $b = ka$ and $a = lb$. Then $a = lka$. If $a = 0$ then $b = k \cdot 0 = 0 = a$. Otherwise $lk = 1$. The only integers whose product is 1 are $k = l = 1$ or $k = l = -1$: if $|k| \geq 2$ then $|kl| \geq 2|l| \neq 1$ unless $l = 0$, which gives $0 = 1$. So $b = a$ or $b = -a$.

S0.5 (i) False: $\sqrt{2} \cdot \sqrt{2} = 2$. (ii) False: $(\sqrt{2})^2 = 2$. (iii) True. Let x be irrational; then $x \neq 0$ (as $0 \in \mathbb{Q}$). Suppose $1/x = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$; then $a \neq 0$ (as $1/x \neq 0$) and $x = \frac{b}{a} \in \mathbb{Q}$, a contradiction.

S0.6 Write $q = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \geq 1$; since $q > 0$, $a \geq 1$. Put $n = b + 1$. Then $\frac{1}{n} = \frac{1}{b+1} < \frac{1}{b} \leq \frac{a}{b} = q$, using $a \geq 1$.

S0.7 Suppose $x^2 - y^2 = 2$, i.e. $(x - y)(x + y) = 2$. Since $(x + y) - (x - y) = 2y$ is even, $x - y$ and $x + y$ are both even or both odd. If both are odd, their product is odd, not 2. If both are even, say $x - y = 2u$, $x + y = 2v$, then $4uv = 2$, so $uv = \frac{1}{2} \notin \mathbb{Z}$. Contradiction.

B CHAPTER 1

S1.1 Contrapositive: if it is not the case that exactly one of a, b is odd, then $a^2 + b^2$ is even. “Not exactly one odd” means both even or both odd. If $a = 2k, b = 2l$: $a^2 + b^2 = 4(k^2 + l^2)$, even. If $a = 2k + 1, b = 2l + 1$: $a^2 + b^2 = 4(k^2 + k + l^2 + l) + 2$, even.

S1.2 (i) False: for $x = 0, xy = 0 \neq 1$ for every y . (ii) True: let $x \in \mathbb{R}$. If $x = 0$ take $y = 0$; the implication is vacuously true. If $x \neq 0$ take $y = 1/x$. (iii) False. Its negation is $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, (x \neq 0 \wedge xy \neq 1)$. Let $y \in \mathbb{R}$. If $y = 0$ take $x = 1$: $xy = 0 \neq 1$. If $y \neq 0$ take $x = 2/y$: $x \neq 0$ and $xy = 2 \neq 1$.

S1.3 $P \Rightarrow (Q \Rightarrow P)$ could only be false with P true and $Q \Rightarrow P$ false; but $Q \Rightarrow P$ is true whenever P is true. $((P \Rightarrow Q) \Rightarrow P) \Rightarrow P$ could only be false with P false and $(P \Rightarrow Q) \Rightarrow P$ true; but if P is false then $P \Rightarrow Q$ is true, so $(P \Rightarrow Q) \Rightarrow P$ is $T \Rightarrow F$, false. $(P \Rightarrow Q) \vee (Q \Rightarrow P)$: if P is false the first disjunct is true; if P is true the second is.

S1.4 (a) Negation: $\exists x \in \mathbb{R}, \forall n \in \mathbb{N}, n \leq x$. The original is true: let $x \in \mathbb{R}$; if $x < 0$ take $n = 0$; otherwise take $n = \lfloor x \rfloor + 1 \in \mathbb{N}$, and $n > x$ since $x < \lfloor x \rfloor + 1$. (b) Negation: $\forall n \in \mathbb{N}, \exists x \in \mathbb{R}, x \geq n$. The original is false: given n , take $x = n$.

S1.5 Let $x \neq y$. By trichotomy, $x < y$ or $y < x$. If $x < y$, put $z = \frac{x+y}{2}$: then $x < z$ since $2x < x + y$, and $z < y$ since $x + y < 2y$. If $y < x$, the same z satisfies $y < z < x$.

S1.6 Contrapositive: if n is even then $n^2 + 2n$ is even. Let $n = 2k$. Then $n^2 + 2n = 4k^2 + 4k = 2(2k^2 + 2k)$.

S1.7 Suppose $\exists x \forall y p(x, y)$ and fix x_0 with $\forall y p(x_0, y)$. Let $y \in X$ be arbitrary. Then $p(x_0, y)$ holds, so $\exists x p(x, y)$ with witness x_0 . Since y was arbitrary, $\forall y \exists x p(x, y)$. The converse fails for $p(x, y)$: “ $x > y$ ” on $\mathbb{Z}$: every y has some $x > y$, but no x exceeds every y .

C CHAPTER 2

S2.1 For any pair (a, b) : $(a, b) \in (X \cup Y) \times Z \Leftrightarrow (a \in X \vee a \in Y) \wedge b \in Z \Leftrightarrow (a \in X \wedge b \in Z) \vee (a \in Y \wedge b \in Z) \Leftrightarrow (a, b) \in X \times Z \vee (a, b) \in Y \times Z \Leftrightarrow (a, b) \in (X \times Z) \cup (Y \times Z)$, using distributivity of $\wedge$ over $\vee$.

S2.2 $(\Rightarrow)$ Suppose $X \subseteq Y$, and suppose $a \in X \setminus Y$. Then $a \in X$, so $a \in Y$, contradicting $a \notin Y$. So $X \setminus Y = \emptyset$. $(\Leftarrow)$ Suppose $X \setminus Y = \emptyset$ and let $a \in X$. If $a \notin Y$ then $a \in X \setminus Y = \emptyset$, impossible. So $a \in Y$.

S2.3 $a \in X \setminus (Y \setminus Z) \Leftrightarrow a \in X \wedge \neg(a \in Y \wedge a \notin Z) \Leftrightarrow a \in X \wedge (a \notin Y \vee a \in Z) \Leftrightarrow (a \in X \wedge a \notin Y) \vee (a \in X \wedge a \in Z) \Leftrightarrow a \in (X \setminus Y) \cup (X \cap Z)$, by De Morgan and distributivity.

S2.4 Disjointness: let $m < n$ in $\mathbb{N}$ and suppose $x \in [m, m+1) \cap [n, n+1)$. Then $x < m+1 \leq n \leq x$, a contradiction. Union: $(\subseteq)$ each $[n, n+1) \subseteq [0, \infty)$ as $n \geq 0$. $(\supseteq)$ let $x \geq 0$ and $n = \lfloor x \rfloor \in \mathbb{N}$; then $n \leq x < n+1$, so $x \in [n, n+1)$.

S2.5 For any set U : $U \in \bigcap_i \mathcal{P}(X_i) \Leftrightarrow \forall i \in I, U \subseteq X_i \Leftrightarrow \forall i \in I, \forall a \in U, a \in X_i \Leftrightarrow \forall a \in U, \forall i \in I, a \in X_i \Leftrightarrow \forall a \in U, a \in \bigcap_i X_i \Leftrightarrow U \subseteq \bigcap_i X_i \Leftrightarrow U \in \mathcal{P}(\bigcap_i X_i)$.

S2.6 For an element a let $c(a)$ be the number of the sets X, Y, Z containing a . Now $a \in X \triangle Y$ iff a lies in exactly one of X, Y . So $a \in (X \triangle Y) \triangle Z$ iff exactly one of “ $a \in X \triangle Y$ ” and “ $a \in Z$ ” holds. If $a \in Z$ this requires a to be in both or neither of X, Y , so $c(a) = 3$ or 1 . If $a \notin Z$ it requires a to be in exactly one of X, Y , so $c(a) = 1$. Thus $a \in (X \triangle Y) \triangle Z$ iff $c(a)$ is odd. The same argument with the roles of X and Z swapped (and $\triangle$ commutative) shows $a \in X \triangle (Y \triangle Z)$ iff $c(a)$ is odd. Hence the sets are equal.

S2.7 (i) False: $X = \emptyset, Y = \{1\}$ gives $(X \setminus Y) \cup Y = \{1\} \neq X$. (In general $(X \setminus Y) \cup Y = X \cup Y$, which equals X iff $Y \subseteq X$.) (ii) True. Note $(X \cup Y) \setminus Y = X \setminus Y$. $(\Rightarrow)$ If $X \setminus Y = X$ and $a \in X \cap Y$, then $a \in X = X \setminus Y$, so $a \notin Y$.

contradiction; so $X \cap Y = \emptyset$. ($\Leftarrow$) If $X \cap Y = \emptyset$: $X \setminus Y \subseteq X$ always, and if $a \in X$ then $a \notin Y$ (else $a \in X \cap Y$), so $a \in X \setminus Y$.

D CHAPTER 3

S3.1 ($\subseteq$) Let $b \in f[f^{-1}[V]]$; fix $a \in f^{-1}[V]$ with $f(a) = b$. Then $b = f(a) \in V$ and $b \in f[X]$. ($\supseteq$) Let $b \in V \cap f[X]$; fix $a \in X$ with $f(a) = b$. Since $f(a) \in V$, $a \in f^{-1}[V]$, so $b \in f[f^{-1}[V]]$.

S3.2 $(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = \text{id}_X$ and $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = \text{id}_Z$. So $f^{-1} \circ g^{-1}$ is an inverse of $g \circ f$, and inverses are unique.

S3.3 ($\Rightarrow$) Let f be injective, $U \subseteq X$, and $b \in f[X \setminus U]$: $b = f(a)$ with $a \notin U$. If $b \in f[U]$ then $b = f(u)$ with $u \in U$, and injectivity gives $a = u \in U$, a contradiction. So $b \in Y \setminus f[U]$. ($\Leftarrow$) Suppose the inclusion holds for every U . Let $f(a) = f(a')$ and suppose $a \neq a'$. Take $U = \{a'\}$. Then $a \in X \setminus U$, so $f(a) \in f[X \setminus U] \subseteq Y \setminus f[U] = Y \setminus \{f(a')\}$, so $f(a) \neq f(a')$: contradiction. Hence $a = a'$.

S3.4 If n is even, $f(n) = n + 1$ is odd, so $f(f(n)) = (n + 1) - 1 = n$. If n is odd, $f(n) = n - 1$ is even, so $f(f(n)) = (n - 1) + 1 = n$. Thus $f \circ f = \text{id}_Z$, so f is its own two-sided inverse, hence a bijection.

S3.5 $g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h$.

S3.6 ($\Rightarrow$) Suppose f surjective and $g \circ f = h \circ f$. Let $b \in Y$; fix a with $f(a) = b$. Then $g(b) = g(f(a)) = h(f(a)) = h(b)$. So $g = h$. ($\Leftarrow$) Contrapositive. Suppose f is not surjective; fix $b \in Y \setminus f[X]$. Define $g, h: Y \rightarrow \{0, 1\}$ by $g(y) = 0$ for all y , and $h(b) = 1$, $h(y) = 0$ for $y \neq b$. For every $a \in X$, $f(a) \neq b$, so $h(f(a)) = 0 = g(f(a))$; thus $g \circ f = h \circ f$. But $g(b) = 0 \neq 1 = h(b)$, so $g \neq h$. Cancellation fails.

S3.7 Let $f: X \rightarrow \mathcal{P}(X)$ and $D = \{a \in X \mid a \notin f(a)\} \in \mathcal{P}(X)$. Suppose f is surjective; fix $d \in X$ with $f(d) = D$. Then $d \in D \Leftrightarrow d \notin f(d) \Leftrightarrow d \notin D$, which is impossible. So f is not surjective.

S3.8 Let $g(x) = a + (b - a)x$. If $0 < x < 1$ then $a < g(x) < b$ as $b - a > 0$, so $g: (0, 1) \rightarrow (a, b)$ is well-defined. Define $k: (a, b) \rightarrow (0, 1)$ by $k(y) = \frac{y-a}{b-a}$; for $a < y < b$ this lies in $(0, 1)$. Then $k(g(x)) = \frac{(b-a)x}{b-a} = x$ and $g(k(y)) = a + (y - a) = y$. So g is a bijection with inverse k .

E CHAPTER 4

S4.1 Base $n = 0$: empty sum $0 = 1! - 1$. Step: $\sum_{k=1}^{n+1} k \cdot k! = (n+1)! - 1 + (n+1)(n+1)! = (n+1)!(n+2) - 1 = (n+2)! - 1$.

S4.2 Base $n = 0$: $6 \mid 0$. Step: $(n+1)^3 - (n+1) = (n^3 - n) + 3n^2 + 3n = (n^3 - n) + 3n(n+1)$. By the IH $n^3 - n = 6k$; and $n(n+1)$ is even (Example 1.3), say $2m$, so $3n(n+1) = 6m$. Hence $(n+1)^3 - (n+1) = 6(k+m)$.

S4.3 For $0 \leq k < n$: $\binom{n}{k} + \binom{n}{k+1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n![(k+1)+(n-k)]}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!} = \binom{n+1}{k+1}$. For $k = n$: $\binom{n}{n} + \binom{n}{n+1} = 1 + 0 = \binom{n+1}{n+1}$; for $k > n$ both sides are 0. Induction on n for $p(n)$: “ $\binom{n}{k} \in \mathbb{N}$ for all $k \in \mathbb{N}$ ”. $p(0)$: $\binom{0}{0} = 1$ and $\binom{0}{k} = 0$ for $k \geq 1$. Step: $\binom{n+1}{0} = 1$, and for $k \geq 0$, $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1} \in \mathbb{N}$ by the IH.

S4.4 Let $p(n)$ be the claim for $2^n \times 2^n$ boards. $p(1)$: a 2×2 board minus one square is exactly one L-piece. Step: given a $2^{n+1} \times 2^{n+1}$ board with one square removed, cut it into four $2^n \times 2^n$ quadrants. The removed square lies in one quadrant. Place one L-piece at the centre of the board covering one square from each of the other three quadrants. Now every quadrant is a $2^n \times 2^n$ board with exactly one square missing; tile each by $p(n)$.

S4.5 Let $p(n)$: no $d \geq 2$ divides both f_n and f_{n+1} . $p(0)$: $f_1 = 1$ has no divisor ≥ 2 . Step: suppose $d \geq 2$ divides f_{n+1} and f_{n+2} . Then $d \mid f_{n+2} - f_{n+1} = f_n$, so d divides both f_n and f_{n+1} , contradicting $p(n)$. Hence $p(n+1)$.

S4.6 Base $n = 1$: $1 \geq 1$. Step: by the IH the sum up to $n+1$ is at least $\sqrt{n} + \frac{1}{\sqrt{n+1}}$, so it suffices that $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$. Multiplying by $\sqrt{n+1} > 0$ this is $\sqrt{n(n+1)} + 1 \geq n+1$, i.e. $\sqrt{n(n+1)} \geq n$, which holds since $n(n+1) \geq n^2$ and both sides are non-negative.

S4.7 Strong induction on $n \geq 1$. Since $2^0 \leq n$ and $2^n > n$, there is a largest $k \in \mathbb{N}$ with $2^k \leq n$. Then $n - 2^k < 2^k$ (otherwise $2^{k+1} \leq n$). If $n - 2^k = 0$, then $n = 2^k$. Otherwise $1 \leq n - 2^k < n$, so by the IH $n - 2^k$ is a sum of distinct powers of 2, each at most $n - 2^k < 2^k$; adding 2^k keeps the powers distinct.

S4.8 Base $n = 1$: $f_1 = 1 = f_3 - 1$. Step: $\sum_{k=1}^{n+1} f_k = (f_{n+2} - 1) + f_{n+1} = f_{n+3} - 1$.

S4.9 Base cases $8 = 3 + 5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$. Step: let $n \geq 11$ and assume the claim for all $8 \leq k < n$. Then $8 \leq n - 3 < n$, so $n - 3 = 3a + 5b$ and $n = 3(a+1) + 5b$. Three base cases are needed because the step reaches back by 3.

F CHAPTER 5

S5.1 Reflexive: $a = 1 \cdot a$. Transitive: Example 0.3-style: $b = ka$, $c = lb$ give $c = (lk)a$. Not symmetric: $1 \mid 2$ but $2 \nmid 1$. Not antisymmetric: $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$.

S5.2 Let $a \in X$; fix b with aRb . By symmetry bRa . By transitivity applied to aRb and bRa , aRa .

S5.3 Reflexive: $x - x = 0 \in \mathbb{Q}$. Symmetric: $y - x = -(x - y) \in \mathbb{Q}$. Transitive: $x - z = (x - y) + (y - z) \in \mathbb{Q}$. $[0] = \{y \mid 0 - y \in \mathbb{Q}\} = \mathbb{Q}$. $[\sqrt{2}] = \{y \mid \sqrt{2} - y \in \mathbb{Q}\} = \{\sqrt{2} - r \mid r \in \mathbb{Q}\} = \{\sqrt{2} + r \mid r \in \mathbb{Q}\}$. If $[\sqrt{2}] = [\sqrt{3}]$ then $r = \sqrt{3} - \sqrt{2} \in \mathbb{Q}$, and $r \neq 0$; then $\sqrt{3} + \sqrt{2} = 1/r \in \mathbb{Q}$, contradicting the Chapter 0 book-level example. So $[\sqrt{2}] \neq [\sqrt{3}]$.

S5.4 Equivalence relations correspond to partitions. Partitions of a 4-element set by block sizes: one block of 4: 1; sizes 3 + 1: 4 (choose the singleton); 2 + 2: 3 (choose the partner of element 1); 2 + 1 + 1: 6 (choose the pair); 1 + 1 + 1 + 1: 1. Total 15.

S5.5 Reflexive: $U \triangle U = \emptyset$ is finite. Symmetric: $U \triangle V = V \triangle U$. Transitive: suppose $U \triangle V$ and $V \triangle W$ are finite. If $a \in U \triangle W$, say $a \in U$, $a \notin W$ (the other case is symmetric): if $a \in V$ then $a \in V \triangle W$; if $a \notin V$ then $a \in U \triangle V$. So $U \triangle W \subseteq (U \triangle V) \cup (V \triangle W)$, a subset of a union of two finite sets, hence finite.

S5.6 $g[X/\sim] = \{g([a]) \mid a \in X\} = \{f(a) \mid a \in X\} = f[X]$. For injectivity: g injective iff for all a, b , $g([a]) = g([b]) \Rightarrow [a] = [b]$, i.e. $f(a) = f(b) \Rightarrow a \sim b$, using $[a] = [b] \Leftrightarrow a \sim b$.

S5.7 $(\Leftarrow)$ Suppose $m \mid n$ and $[a]_n = [a']_n$. Then $n \mid a - a'$, and $m \mid n$, so $m \mid a - a'$ by transitivity of $\mid$; thus $[a]_m = [a']_m$. $(\Rightarrow)$ If the rule is well-defined, then since $[0]_n = [n]_n$ we must have $[0]_m = [n]_m$, i.e. $m \mid n - 0 = n$.

S5.8 $(\Rightarrow)$ An equivalence relation is reflexive; and if aRb and aRc , then bRa by symmetry, and bRa , aRc give bRc by transitivity. $(\Leftarrow)$ Suppose R is reflexive and has the stated property. Symmetric: let aRb . Also aRa . Applying the property to aRb and aRa gives bRa . Transitive: let aRb and bRc . By symmetry bRa . Applying the property to bRa and bRc gives aRc .